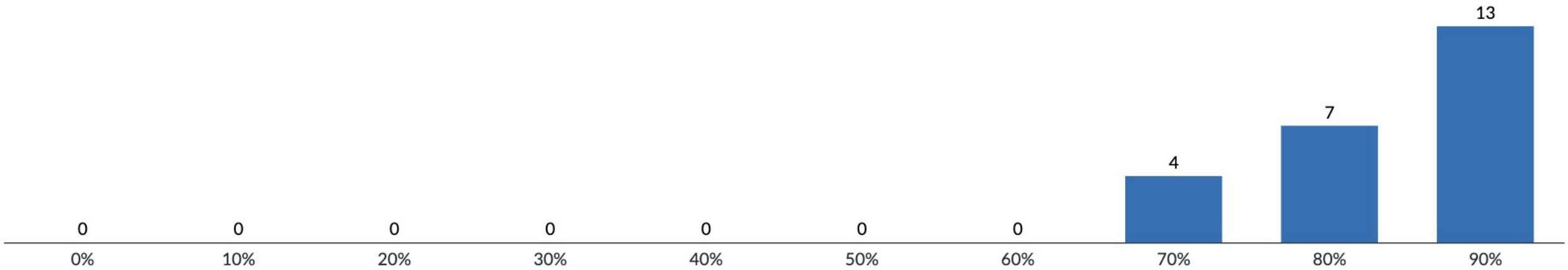
CS-229

- Quiz 1
- Intro Computer Vision
  - Tasks
  - Architectures / backbones / prior knowledge

Prompt: “A robot is doing taxes, but is in despair because it's too complicated.”

100%	78%	93%	100%	06:19:21
High Score	Low Score	Mean Score	Median Score	Mean Elapsed Time

Score Distribution



Get some scratch paper ready for the next few.
Suppose we have three words, "Attention to words", and they have word embeddings, $x_{\text{Attention}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, x_{\text{to}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, x_{\text{words}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. For simplicity, assume that the key, query and value matrices for each word are all equal,
$k_i = \mathbb{I}\, x_i, q_i = \mathbb{I}\, x_i, v_i = \mathbb{I}\, x_i$.
We're going to calculate the output of a dot-product attention on this input. (Don't use scaling, so we get softmax(QK^T) V, where Q, K, V are matrices that stack the vectors, like $Q = \begin{pmatrix} - & \vec{q}_1 & - \\ - & \vec{q}_2 & - \\ - & \vec{q}_3 & - \end{pmatrix}$).
First we need the dot product matrix, then we take the softmax. The output of the second position (with "to" as the input) pays the most "attention" (i.e. has highest probability) to which token?

Answer Frequency Summary

	Answer	Respondents	Percentage
✓	words	19	79%
×	to	3	13%

Continuing what is the output vector in the first position (associated with "Attention")
(x_i refers to vectors from the previous question.)

Answer Frequency Summary

	Answer	Respondents	Percentage
×	1/2 xAttention + 1/2 xwords	3	13%
✓	0.42 xAttention + 0.16 xto + 0.42 xwords	18	75%
×	1/8 xAttention + 1/4 xto + 5/8 xwords	0	0%
×	xAttention	3	13%

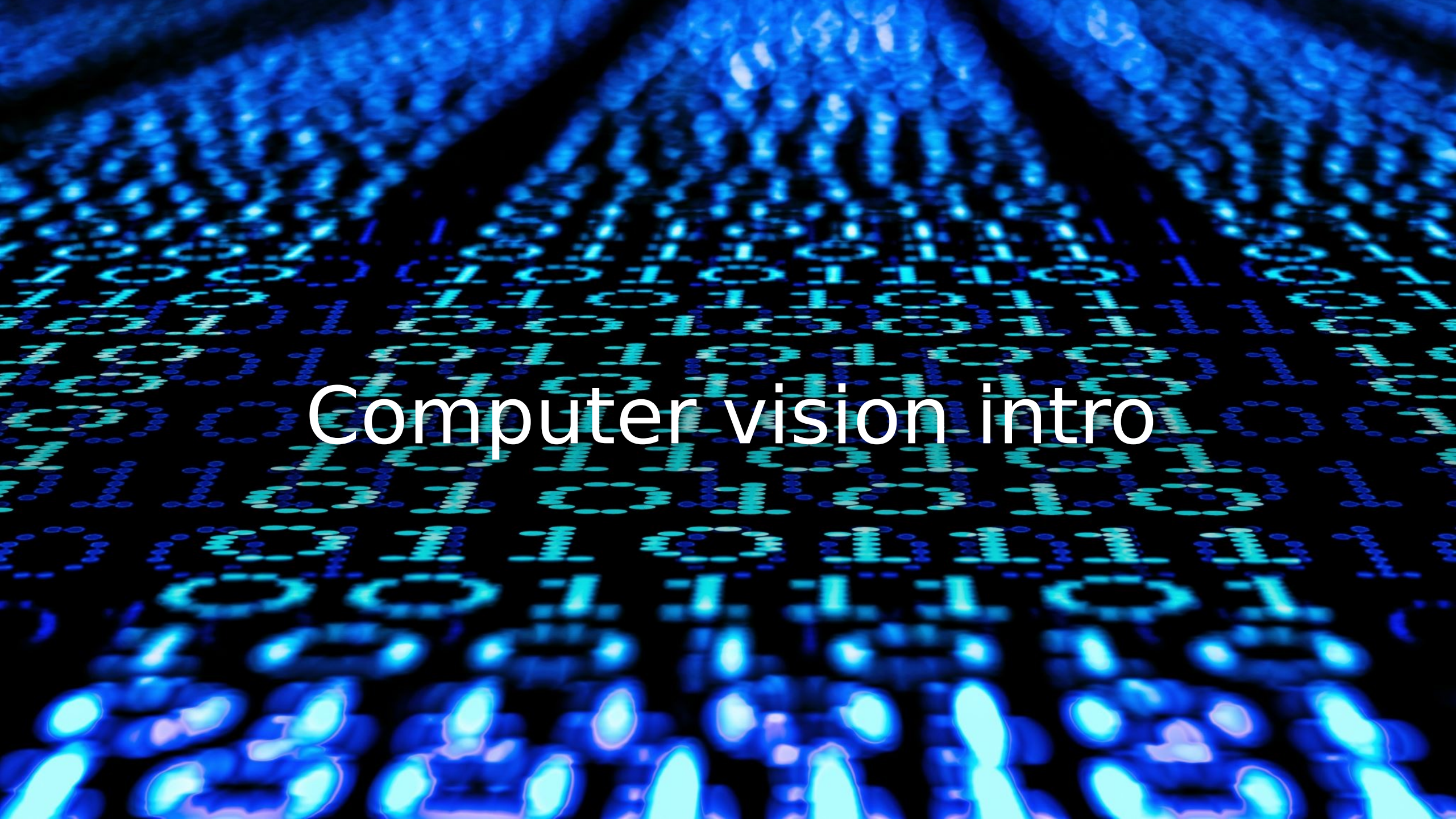
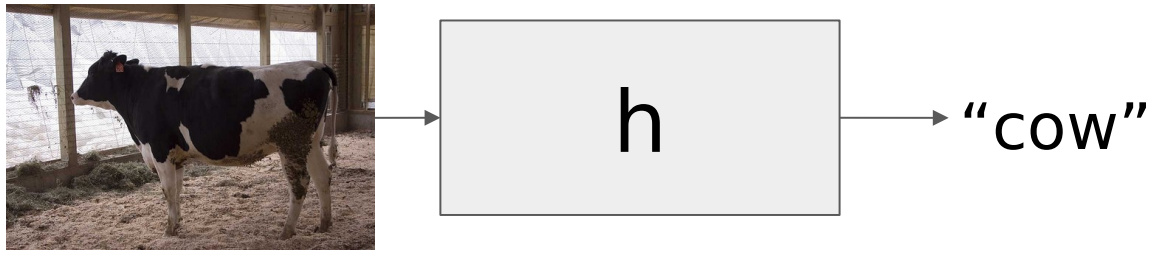
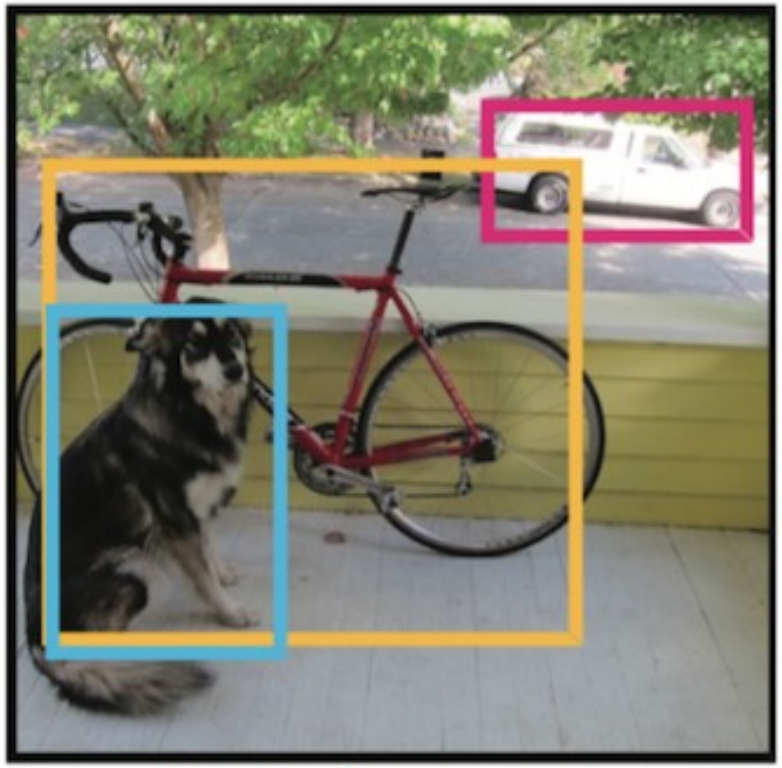

Computer vision intro

Image Classification



Object detection

- Output a bounding box and a category
- Many objects possible per image



https://lilianweng.github.io/posts/2018-12-27-object-recognition-part-4/

(Semantic) Segmentation – assign a category to each *pixel*

Segment Anything

Instance segmentation

Segment Anything

Many other tasks

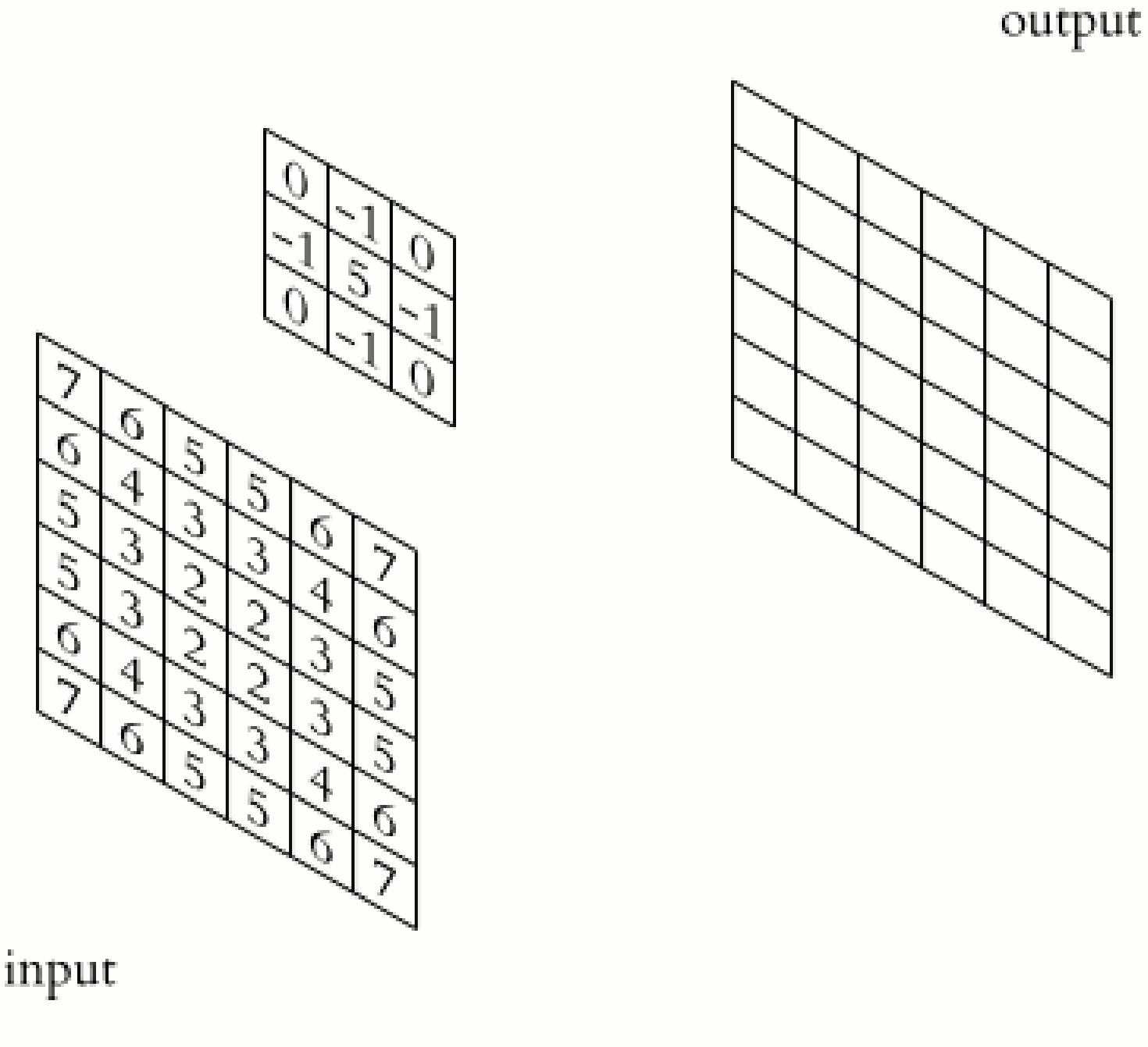
- Face or pose recognition
- 3D reconstruction or depth mapping
- Video tasks
  - Object tracking
  - Activity recognition
  - Temporal segmentation
- Multimodal (A little on Monday)
  - Image captioning
  - Visual question answering
  - OCR – optical character recognition
- Generative tasks (later in the course)
  - Image restoration
  - Inpainting
  - Super-resolution
  - Style transfer / image synthesis

Architectures encode our intuition / prior knowledge about vision

- Convolution – locality and spatial invariance
- Pooling – translation invariance
- Attention

Convolutio

- Convolutional kernel – combines each pixel with info from neighboring pixels (locality)
- Same weights applied spatially everywhere (spatial invariance)
- If there are N pixels, this is a N-> N mapping. How many weights would it generally take?



Convolution* in 1D (invariant/equivariant?)

- Input vector **x:**

$$\mathbf{x} = [x_1, x_2, \ldots, x_I]$$

- Output is weighted sum of neighbors:

$$z_i = \omega_1 x_{i-1} + \omega_2 x_i + \omega_3 x_{i+1}$$

- Convolutional kernel or filter:

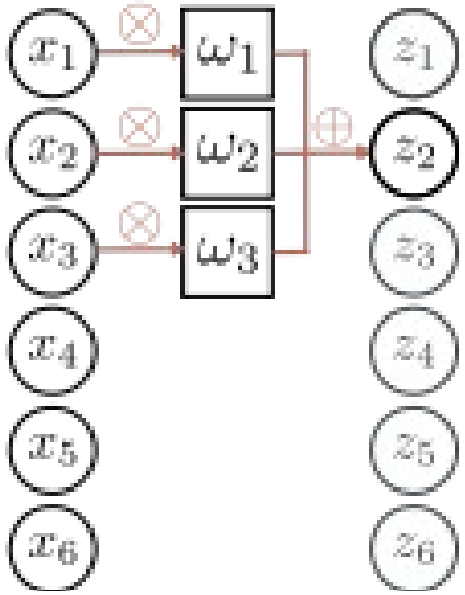
$$\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^T$$

Kernel size = 3

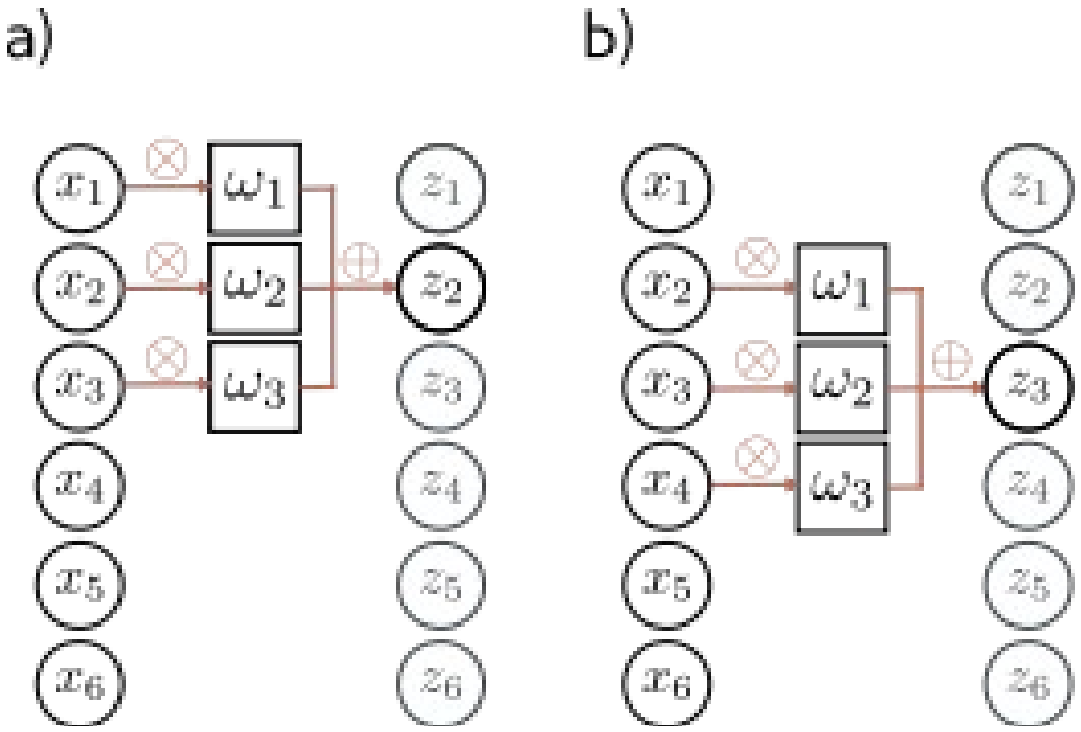
* different from signal processing

Convolution with kernel size 3

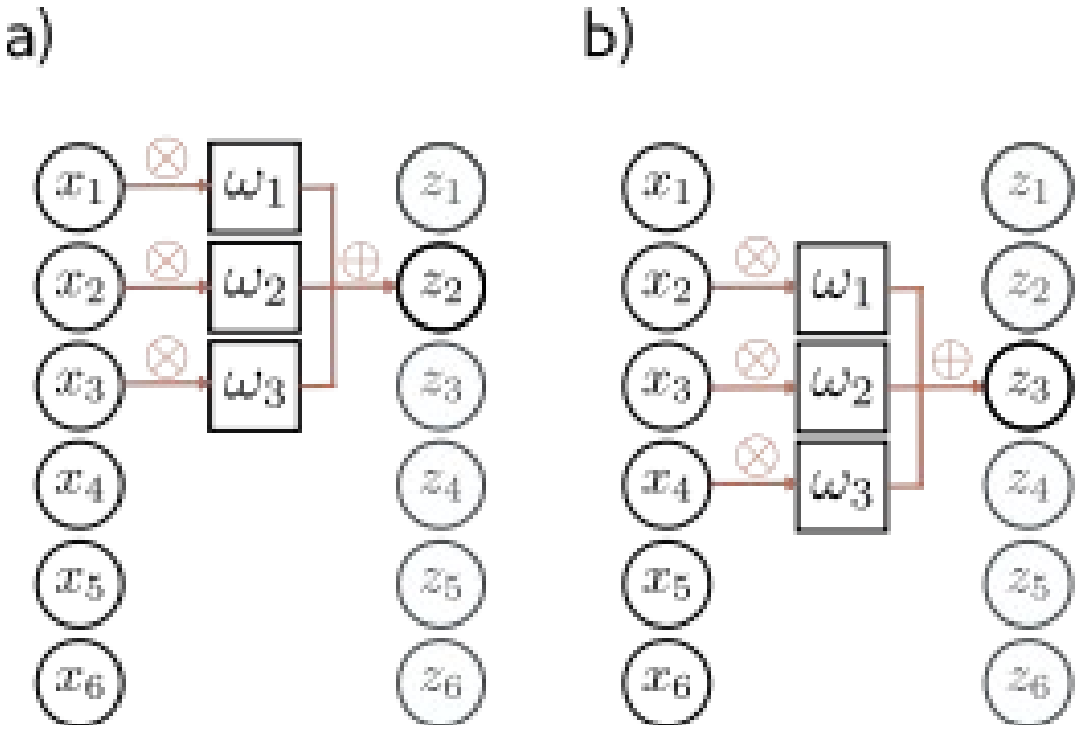
a)



Convolution with kernel size 3



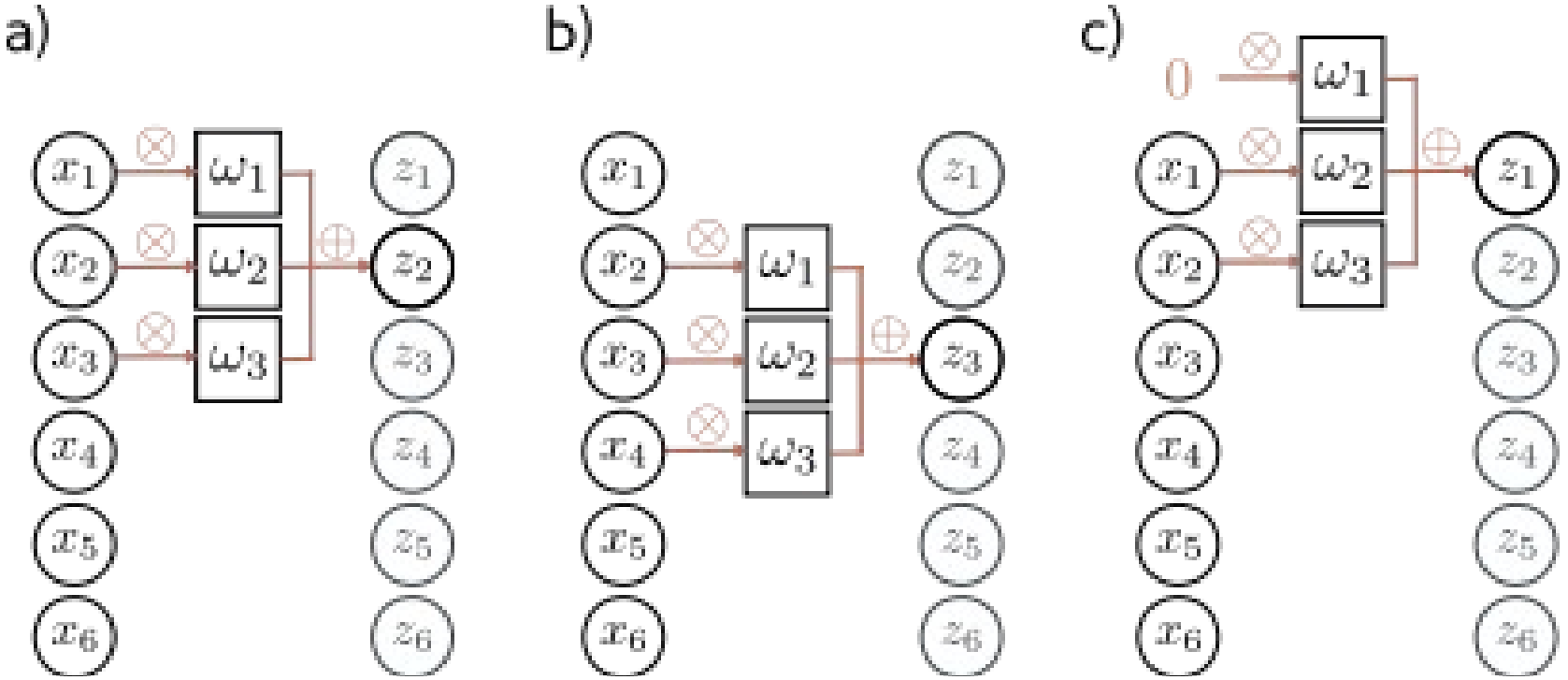
Convolution with kernel size 3



Equivariant to translation of input

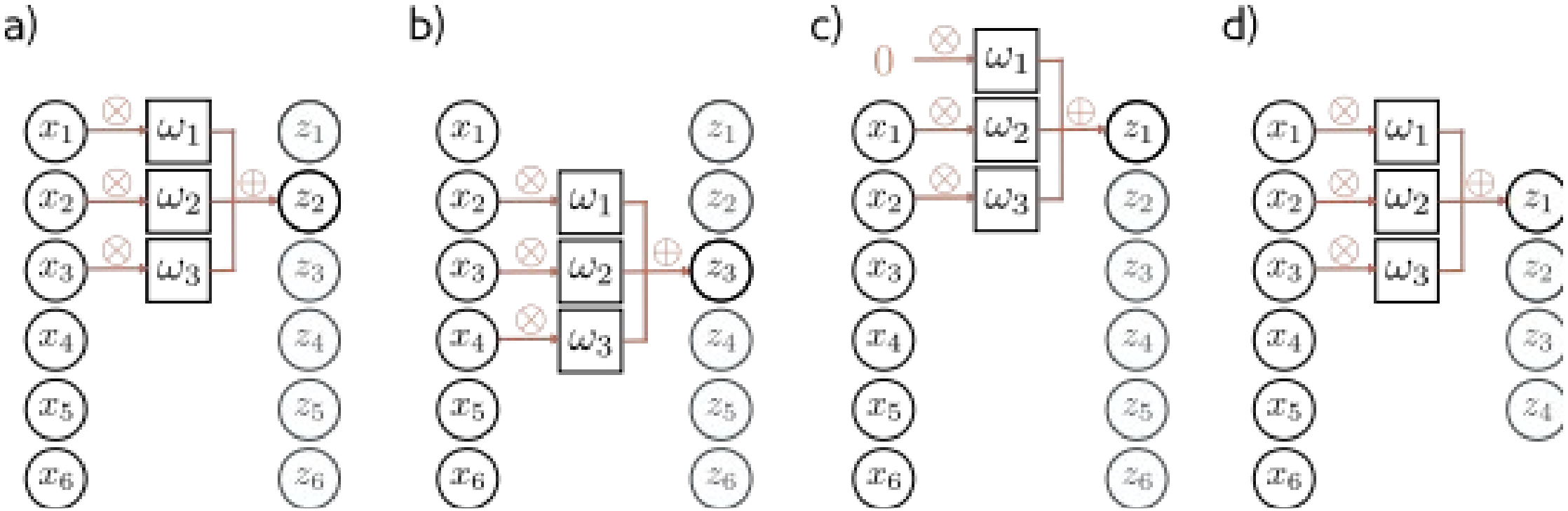
$$\mathbf{f}[\mathbf{t}[\mathbf{x}]] = \mathbf{t}[\mathbf{f}[\mathbf{x}]]$$

Zero padding



Treat positions that are beyond end of the input as zero.

“Valid” convolutions

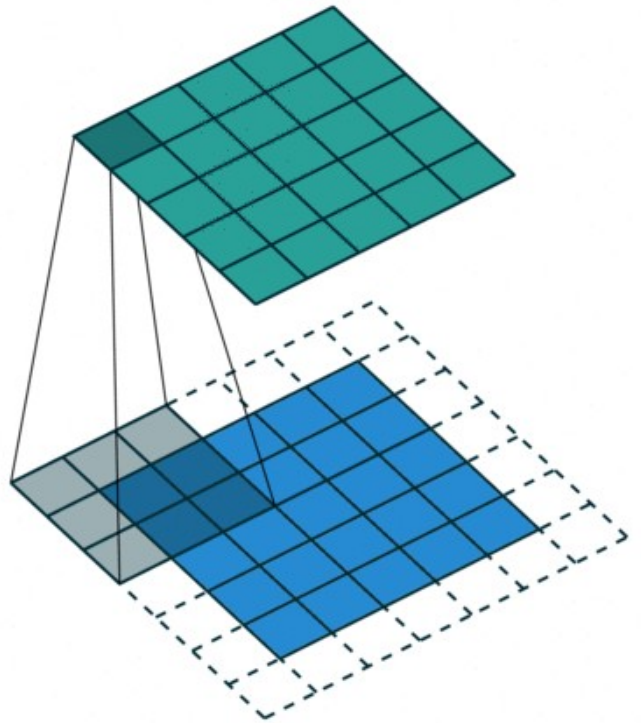


Only process positions where kernel falls in image (smaller output).

Padding in convolution (2d)

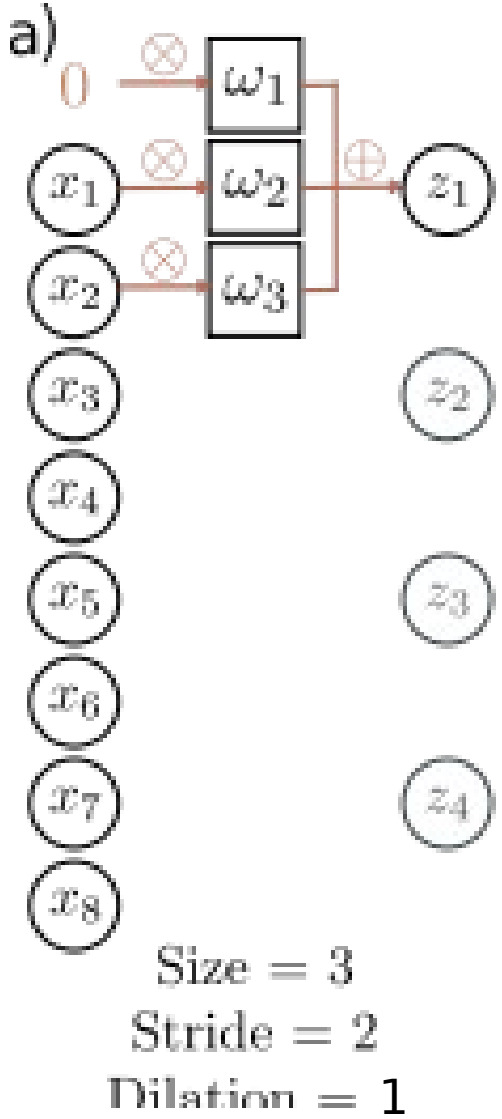
https://hannibunny.github.io/mlbook/neuralnetworks/convolutionDemos.html

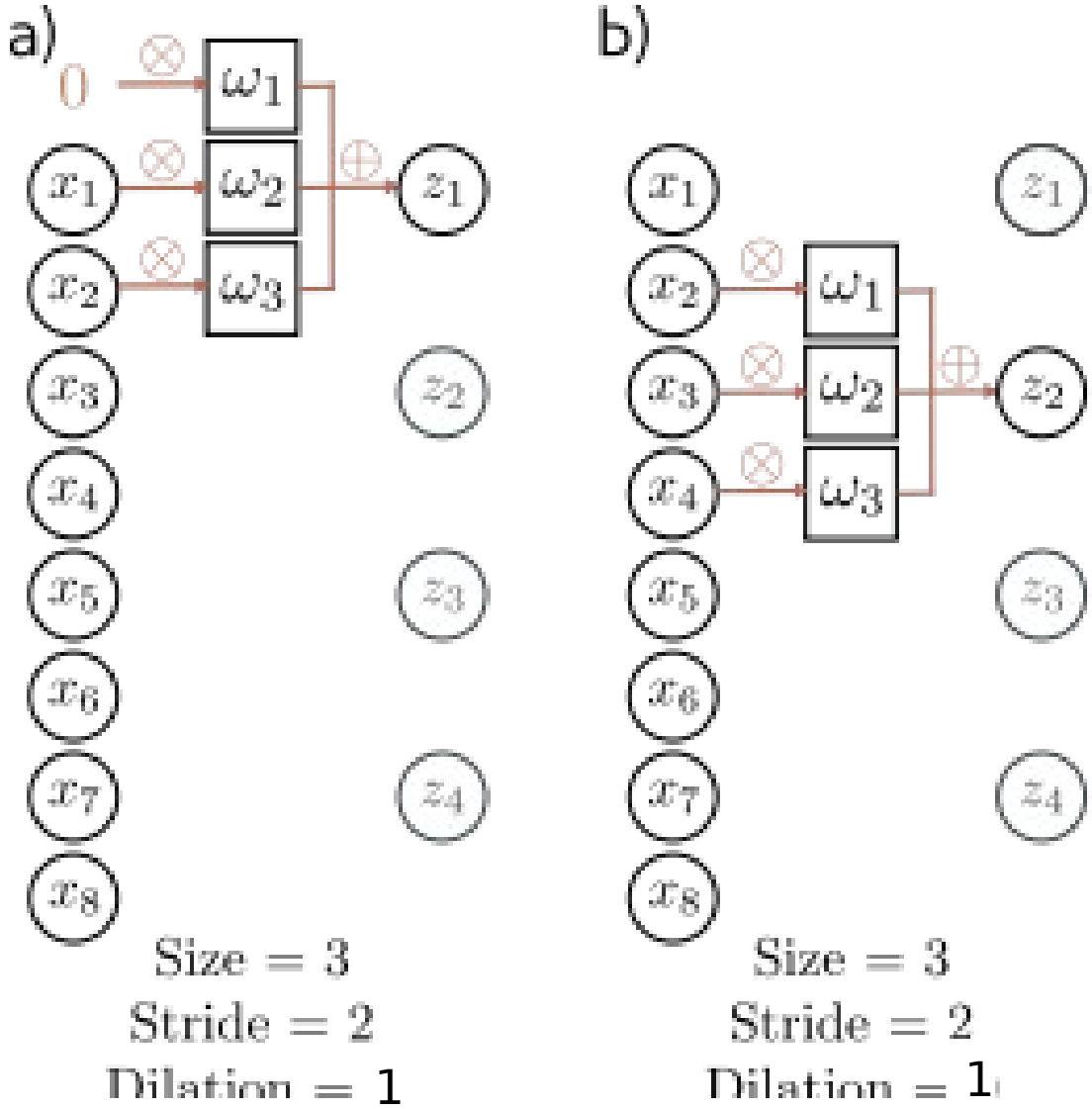
For me visualizations make this easie

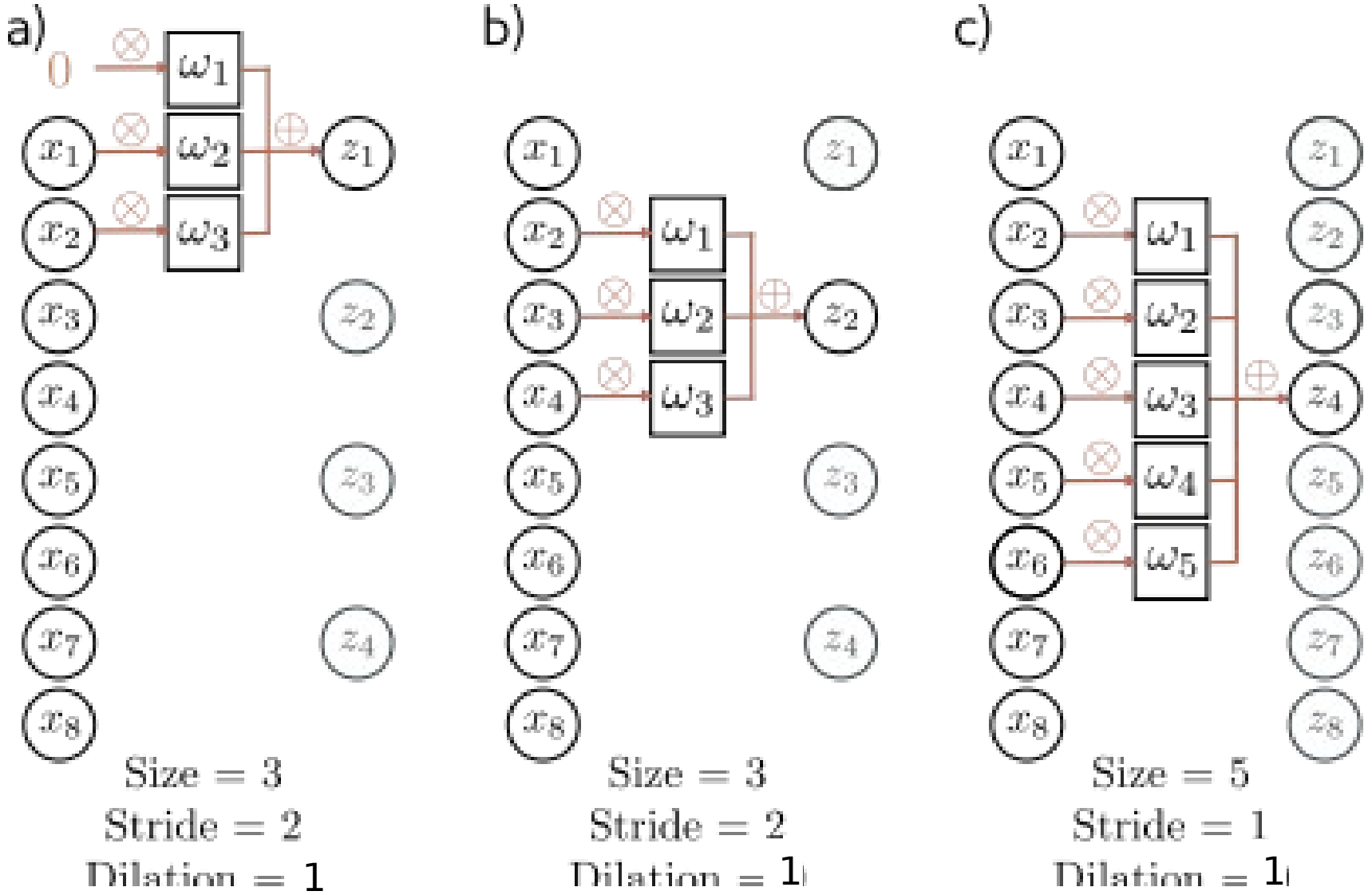


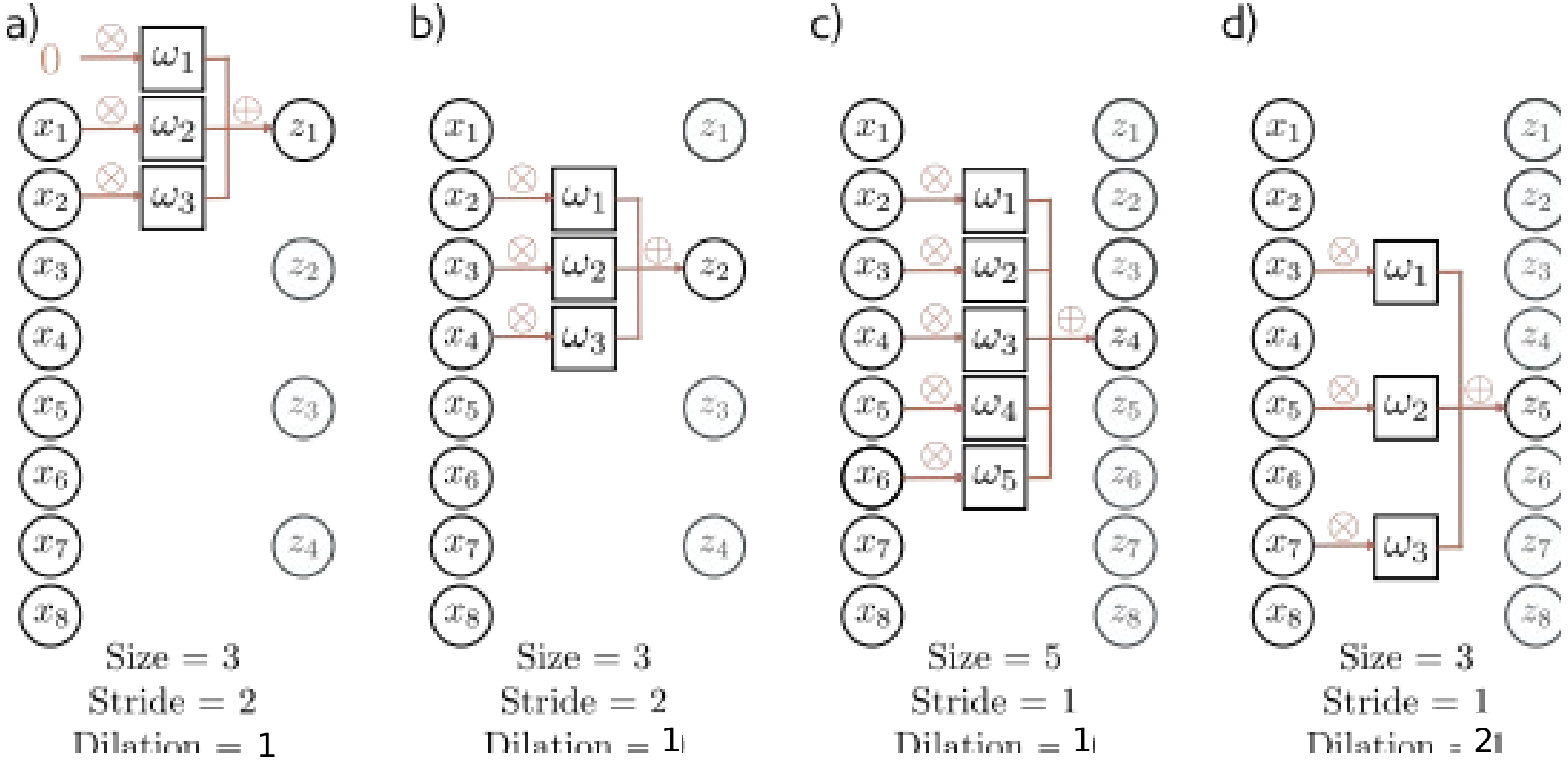
Stride, kernel size, and dilation

- Stride = shift by k positions for each output
  - Decreases size of output relative to input
- Kernel size = weight a different number of inputs for each output
  - Combine information from a larger area
  - But kernel size 5 uses 5 parameters
- Dilated or atrous convolutions = intersperse kernel values with zeros
  - Combine information from a larger area
  - Fewer parameters

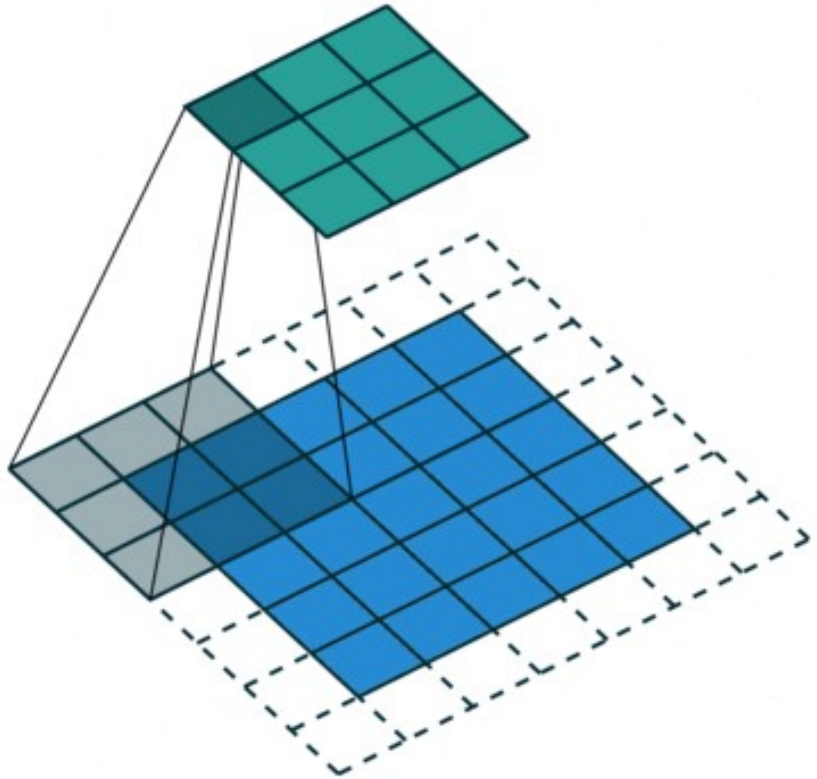

a)
0
ω_1
ω_2
ω_3
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
z_1
z_2
z_3
z_4
Size = 3
Stride = 2
Dilation = 1


a)
0
ω_1
ω_2
ω_3
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
z_1
z_2
z_3
z_4
Size = 3
Stride = 2
Dilation = 1
b)
ω_1
ω_2
ω_3
x_1
x_2
x_3
x_4
x_5
x_6
x_7
x_8
z_1
z_2
z_3
z_4
Size = 3
Stride = 2
Dilation = 1

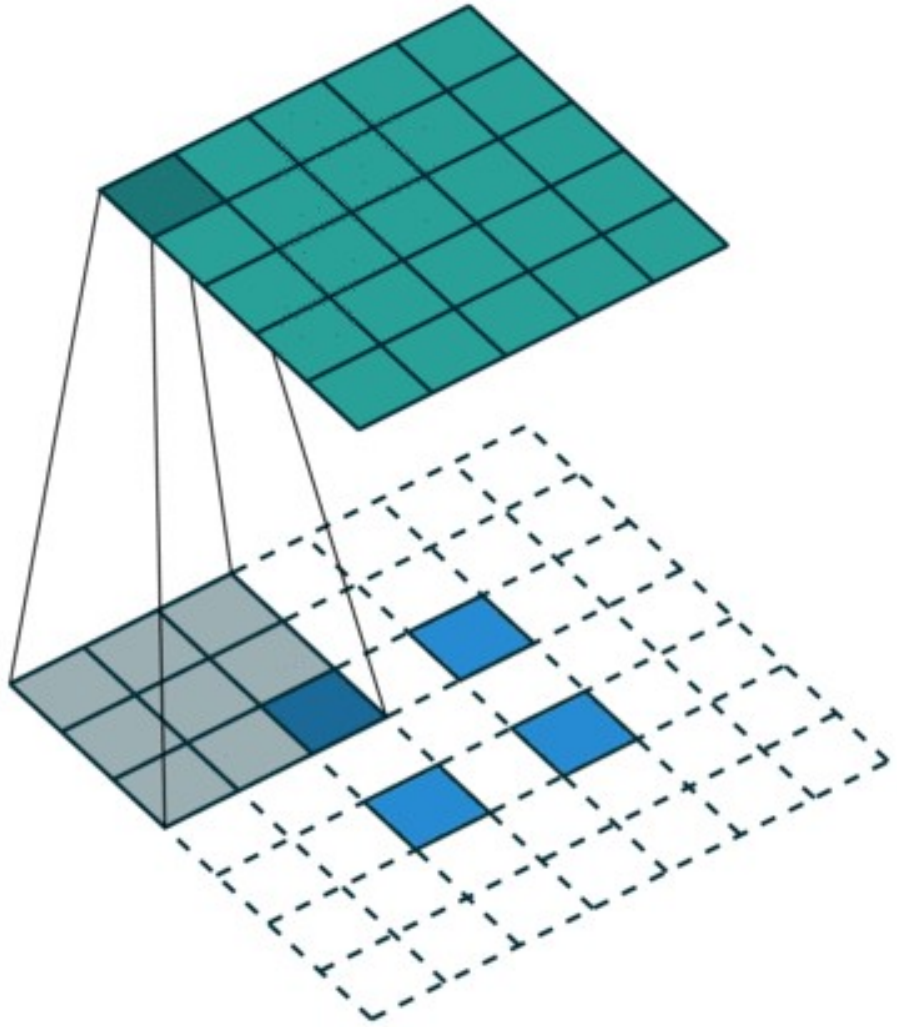

a)
b)
c)
Size = 3
Stride = 2
Dilation = 1
Size = 3
Stride = 2
Dilation = 1
Size = 5
Stride = 1
Dilation = 1


a)
b)
c)
d)
Size = 3
Stride = 2
Dilation = 1
Size = 3
Stride = 2
Dilation = 1
Size = 5
Stride = 1
Dilation = 1
Size = 3
Stride = 1
Dilation = 2

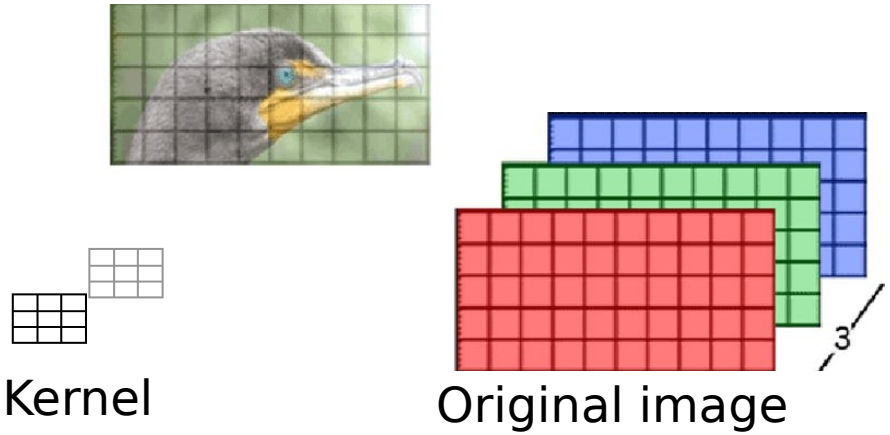
Strided convolution – apply to every other pixel – output is half as big



Convolution to upscale an image



Convolution equation



Kernel

Original image

Output “image”

$$I'_{x,y,k} = \sum_{i=-K}^{K} \sum_{j=-K}^{K} \sum_{c=1}^{C} K_{i,j,c,k} \cdot I_{x+i,y+j,c}$$

$I_{x,y,c}$ - value of input image at pixel location (x, y) and channel c.

I' is the output

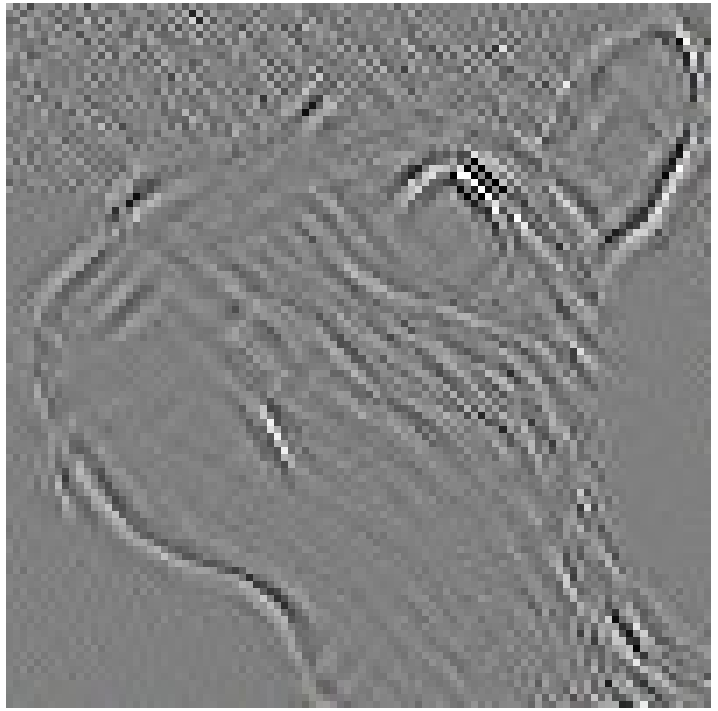
$K_{i,j,c,k}$ is the weight of the convolution kernel at position (i, j) for input channel c and output channel k.

C is the number of input channels.

K is the size of the kernel (assuming a square kernel of size $(2K+1) \times (2K+1)$).

What do convolutions learn?

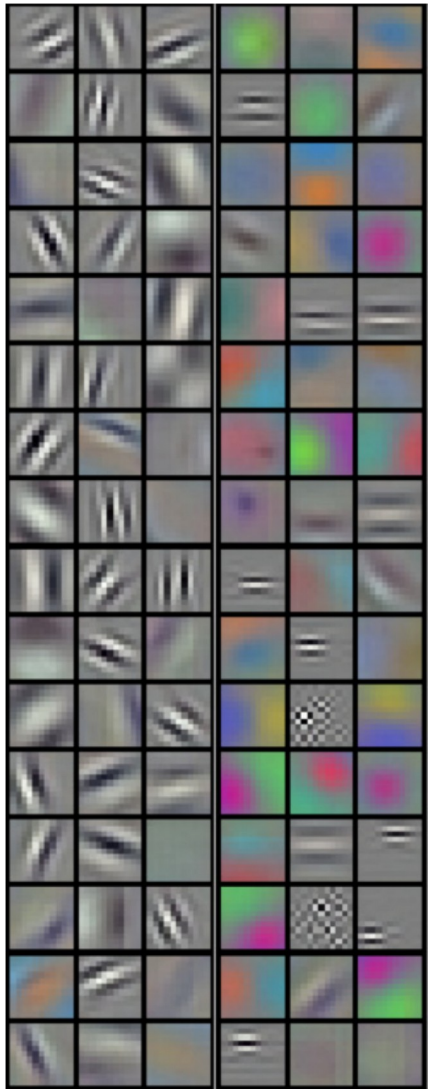
$$* \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} =$$

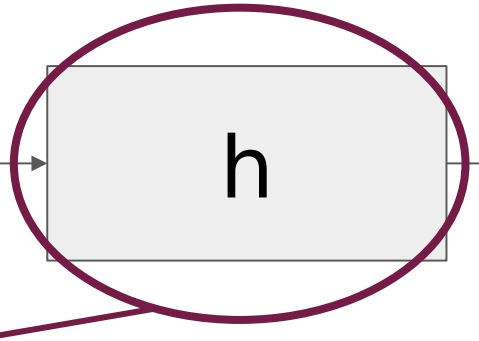


Weird example from Wikipedia: lots of different image features are possible

What is the “receptive field” of a pixel on the right?

What do convolutions at each layer learn?





“cow”

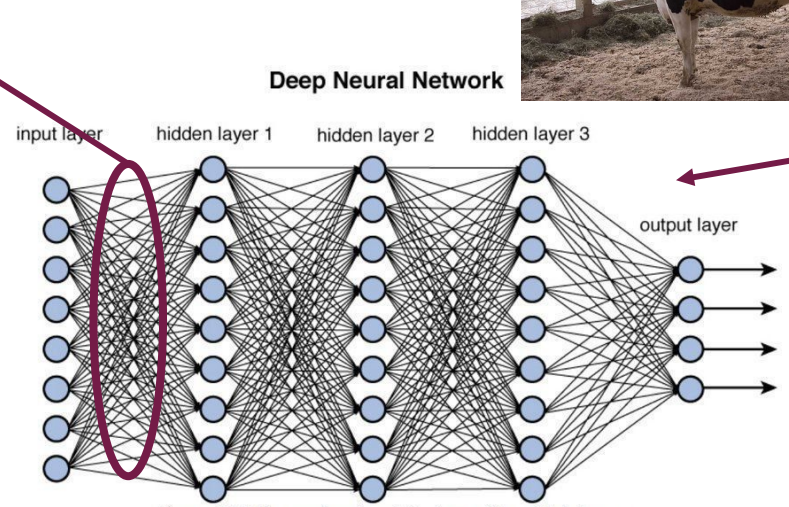
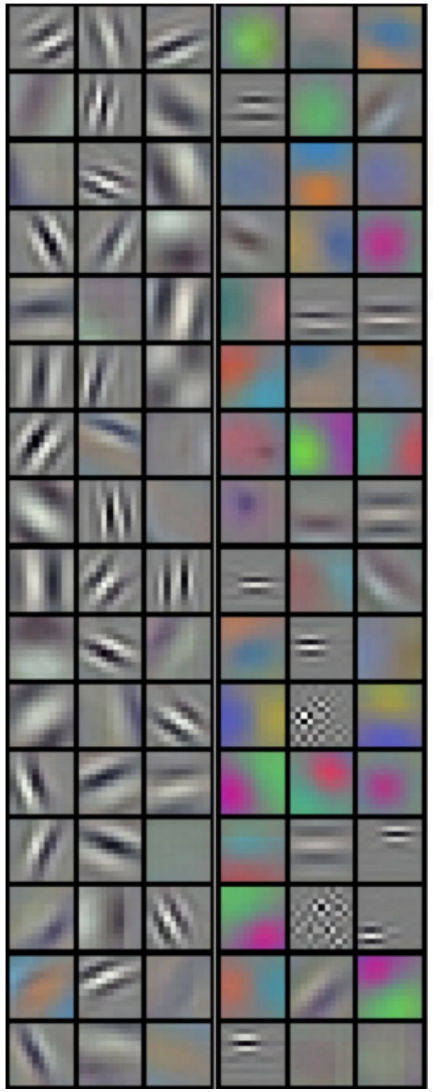
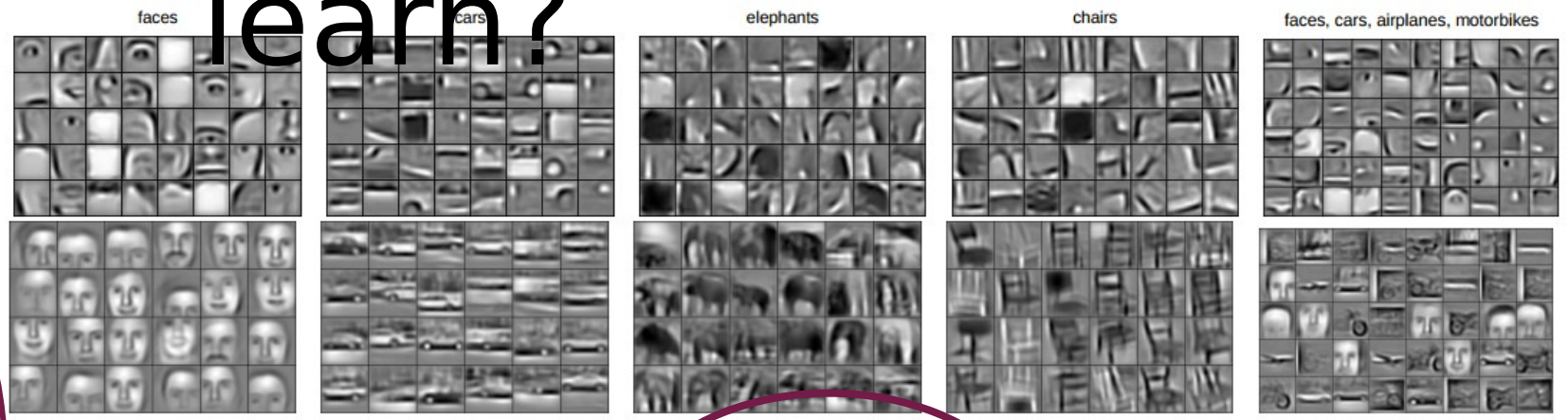


Figure 12.2 Deep network architecture with multiple layers.

What do convolutions at each layer learn?





h

“cow”

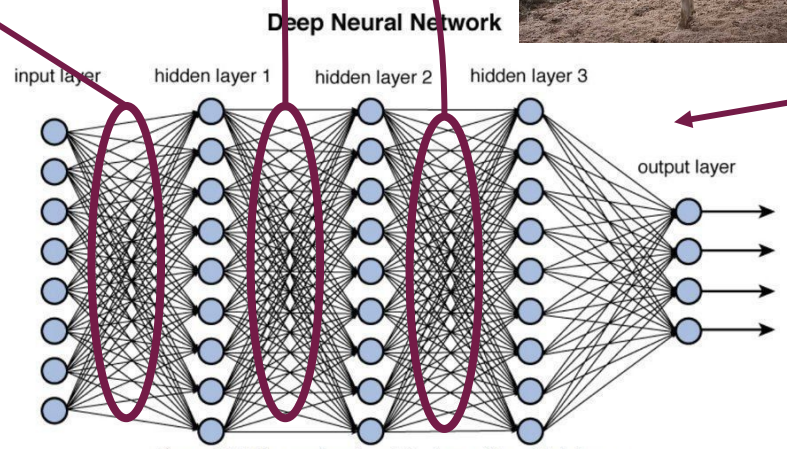
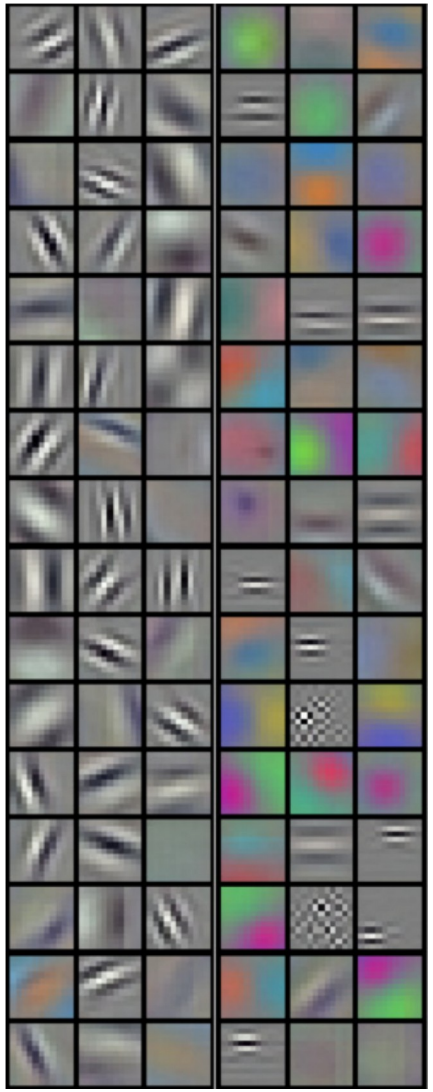
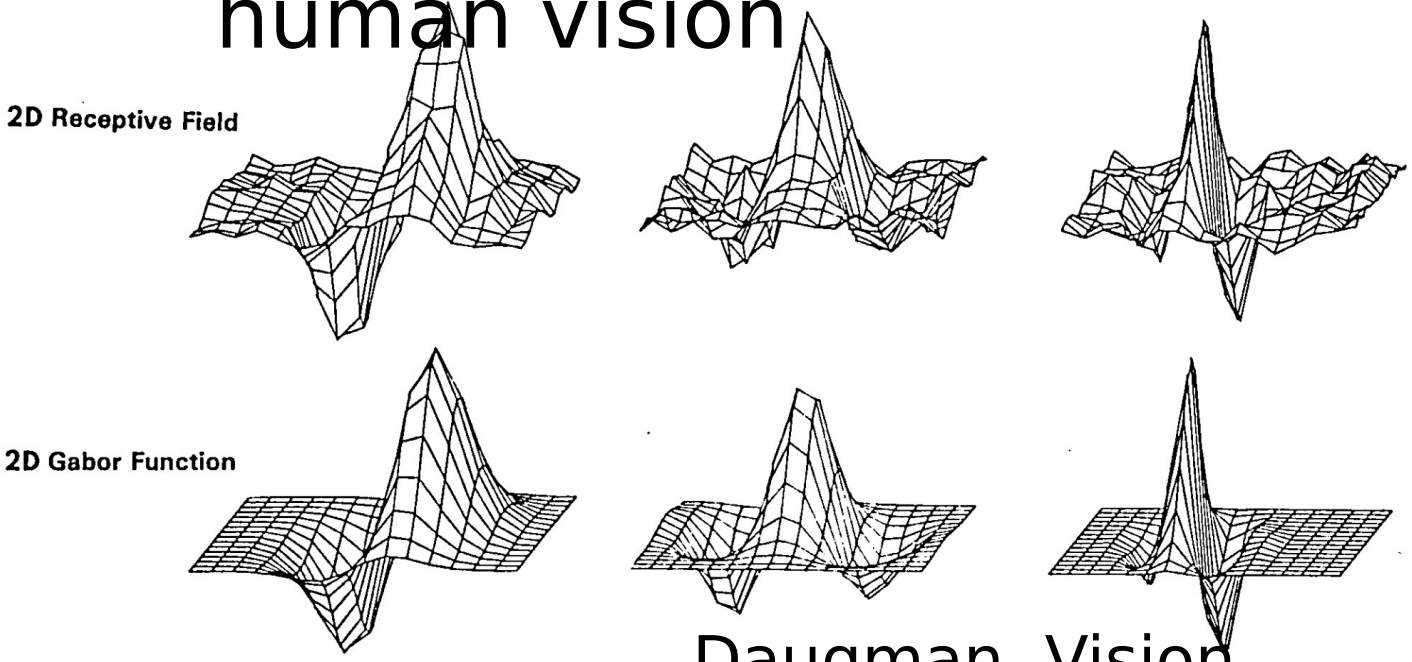


Figure 12.2 Deep network architecture with multiple layers.

Are these familiar?

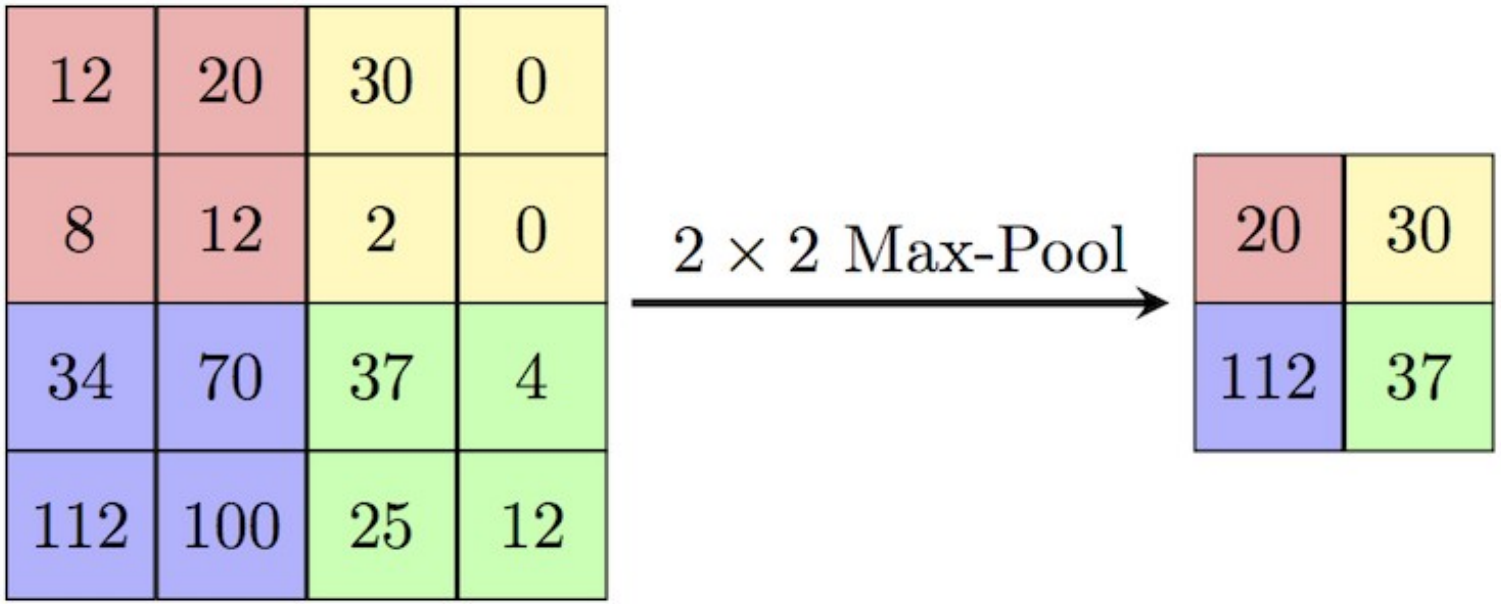


Neuroscientists discovered them in the 80s (called Gabor Filters) as fundamental to human vision



Daugman, Vision Research,1980

Max pooling adds some robustness to translations



Ways to reduce dimensionality of image

Max pooling

- no parameters
- Simpler / cheaper operation
- Translation invariance

Strided convolutions

- Learnable parameters
- More flexible

Fully-Convolutional Networks (FCN)

Uses Convolutions then deconvolutions (https://arxiv.org/abs/1411.4038)

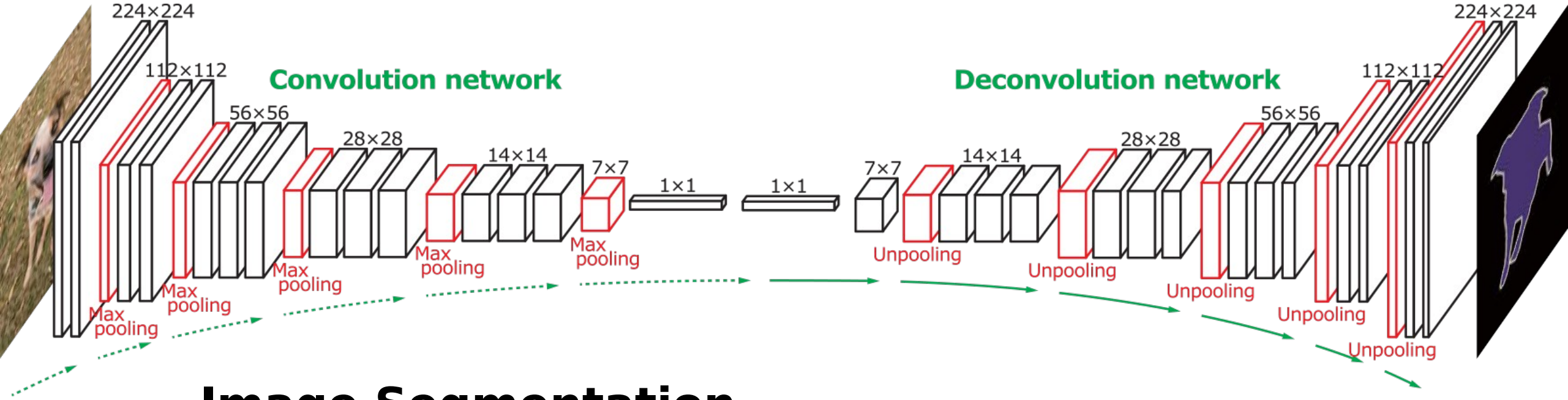


Image Segmentation

U-Net

Like FCNs, but adds skip connections (paper)

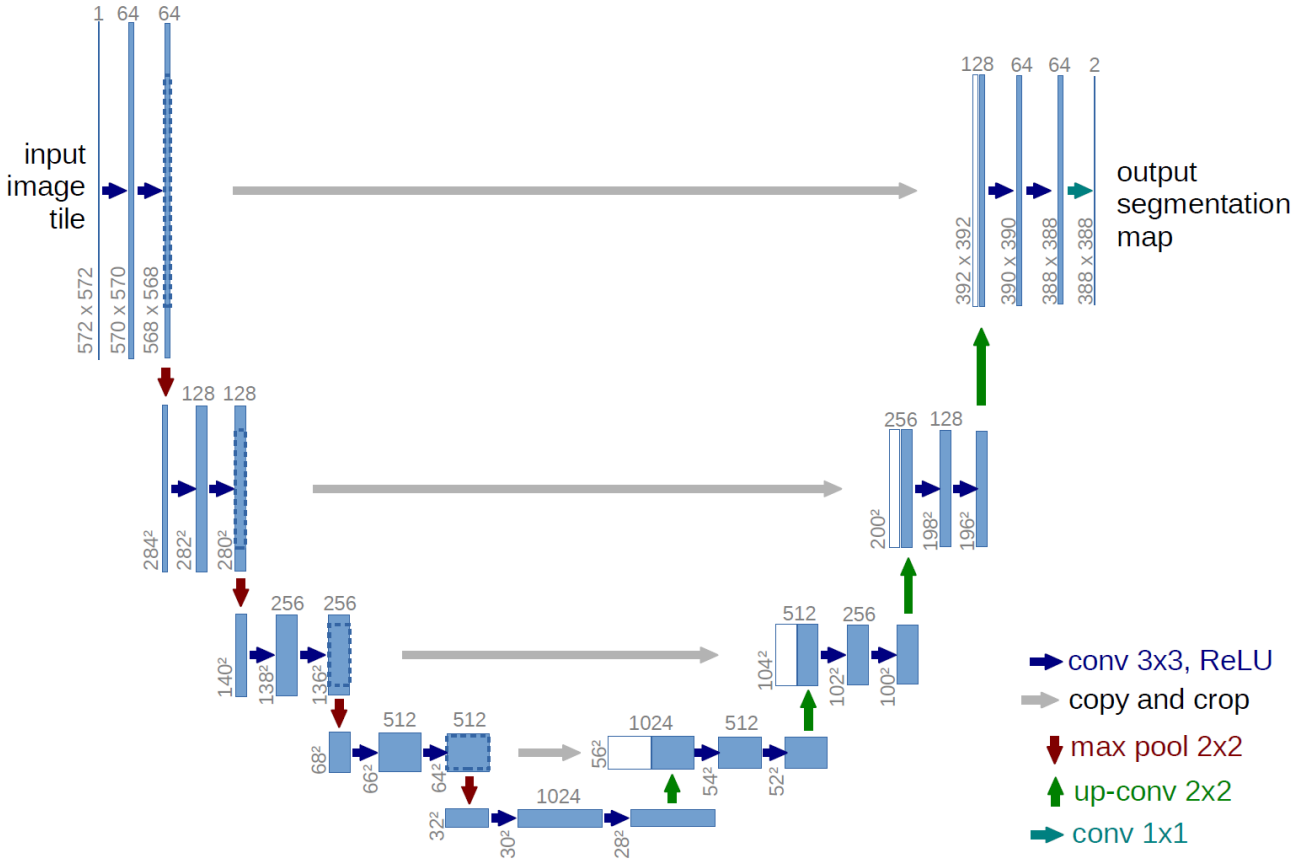
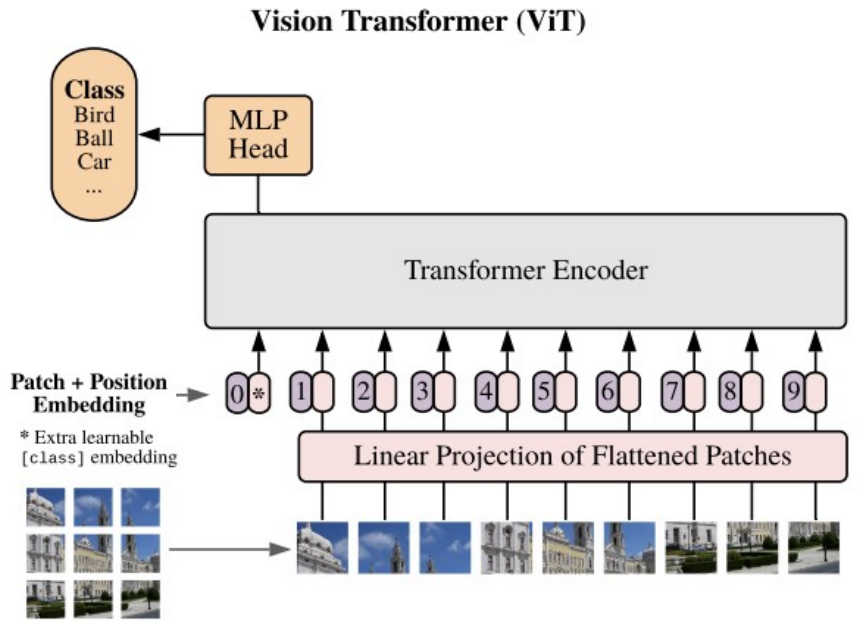


Image Segmentation

Image generation (diffusion – later in course)

Abbreviated history of vision “backbones”

- Convolutional
  - VGG
  - Resnets, WideResNets
  - EfficientNet
- Vision transformer
  - Original
  - Swin transformer – hierarchical version ConvNext (early 2022)
- Combine conv + attention: MaxViT (late 2022)
- 2026? Attn more common, but arch less important than scale/training (MONDAY)



Residual Neural Networks (ResN

Composed of Residual blocks.

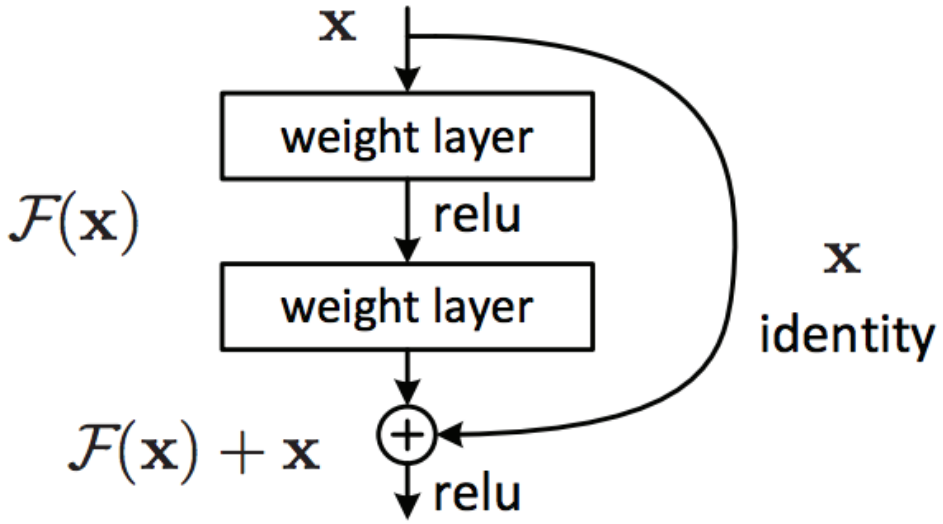
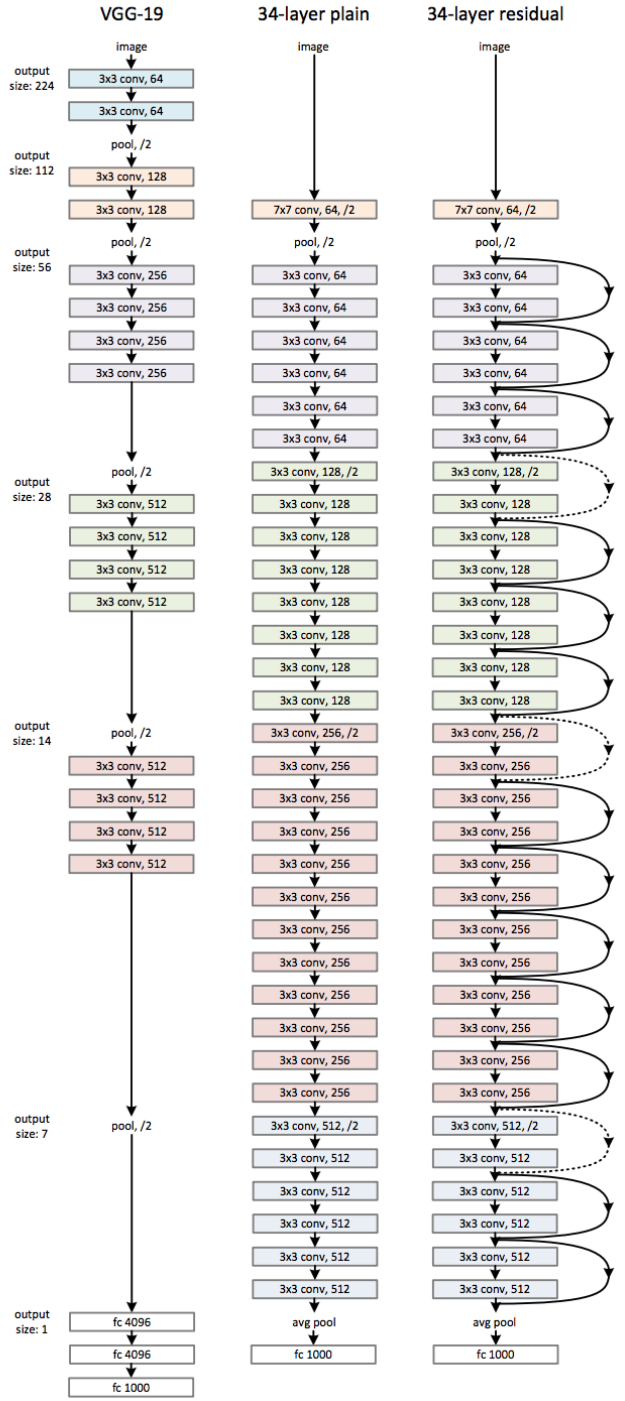
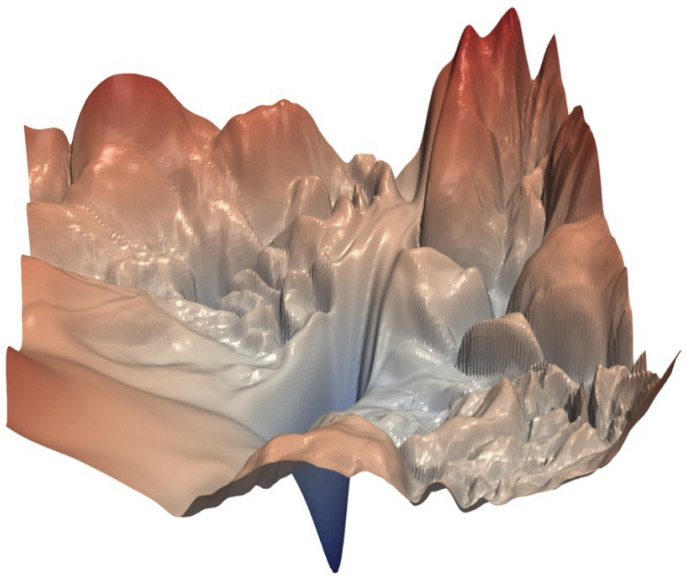


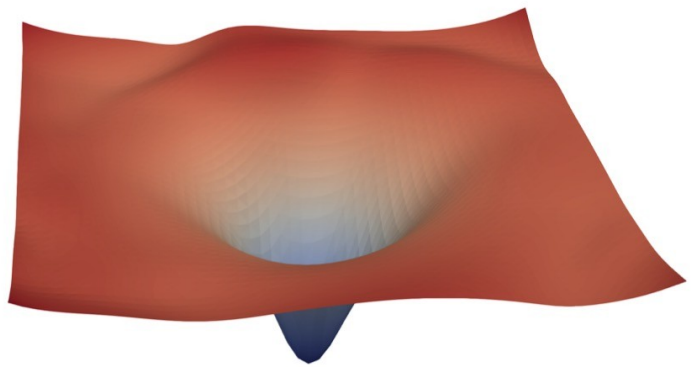
Figure 2. Residual learning: a building block.



Visualizing the Loss Landscape of Neural Nets



(a) without skip connections



(b) with skip connections

How to grow your (residual) neural net

EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks

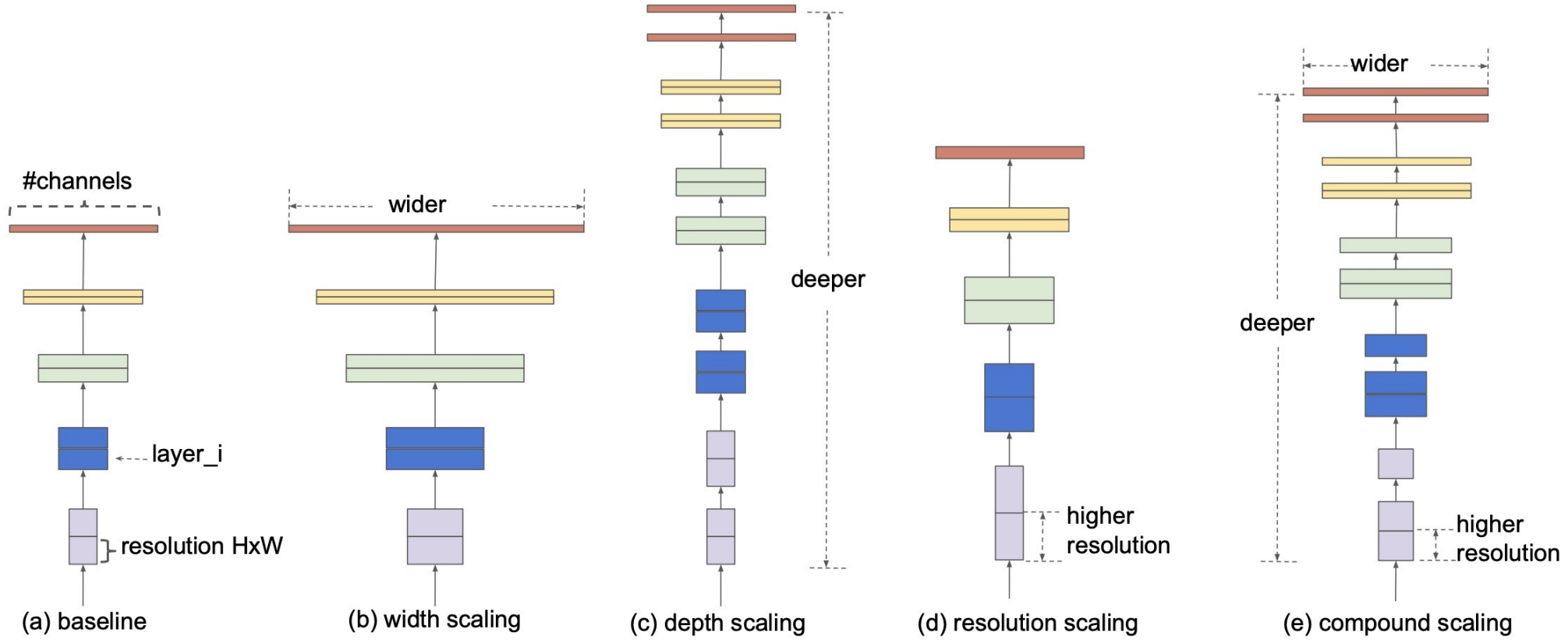


Figure 2. **Model Scaling.** (a) is a baseline network example; (b)-(d) are conventional scaling that only increases one dimension of network width, depth, or resolution. (e) is our proposed compound scaling method that uniformly scales all three dimensions with a fixed ratio.

The Vision Transformer

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]

[*]equal technical contribution, [†]equal advising

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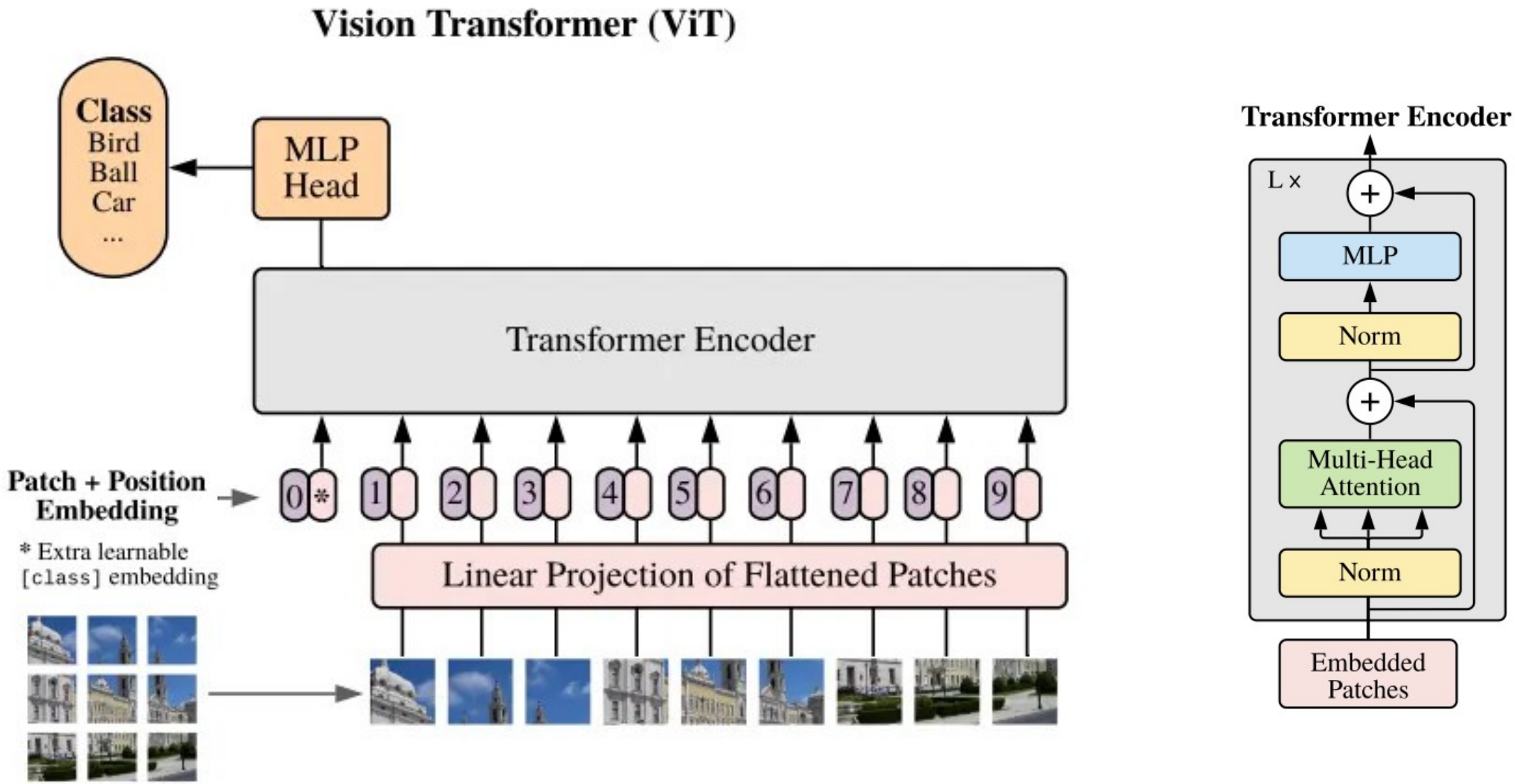
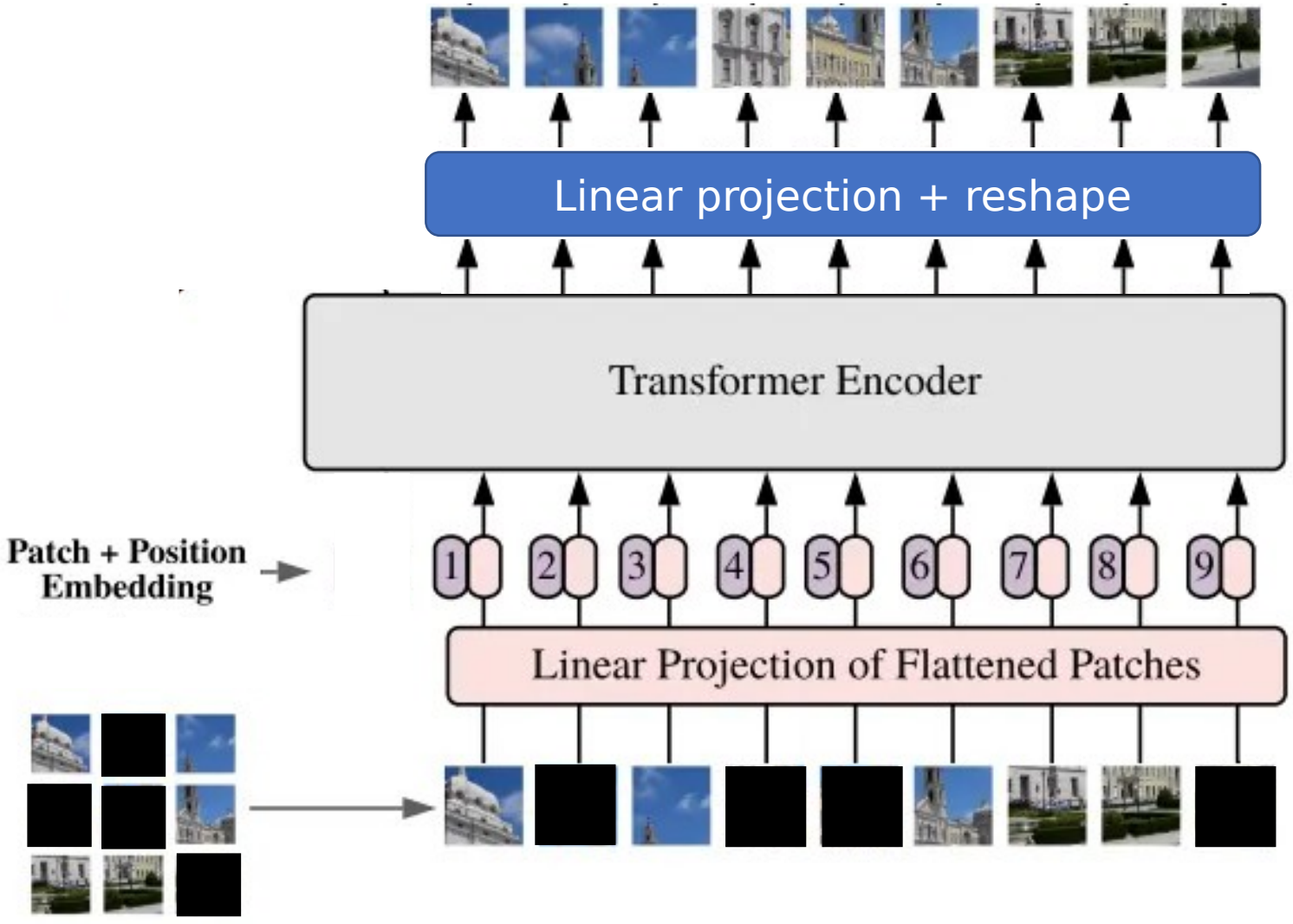
ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.[1]

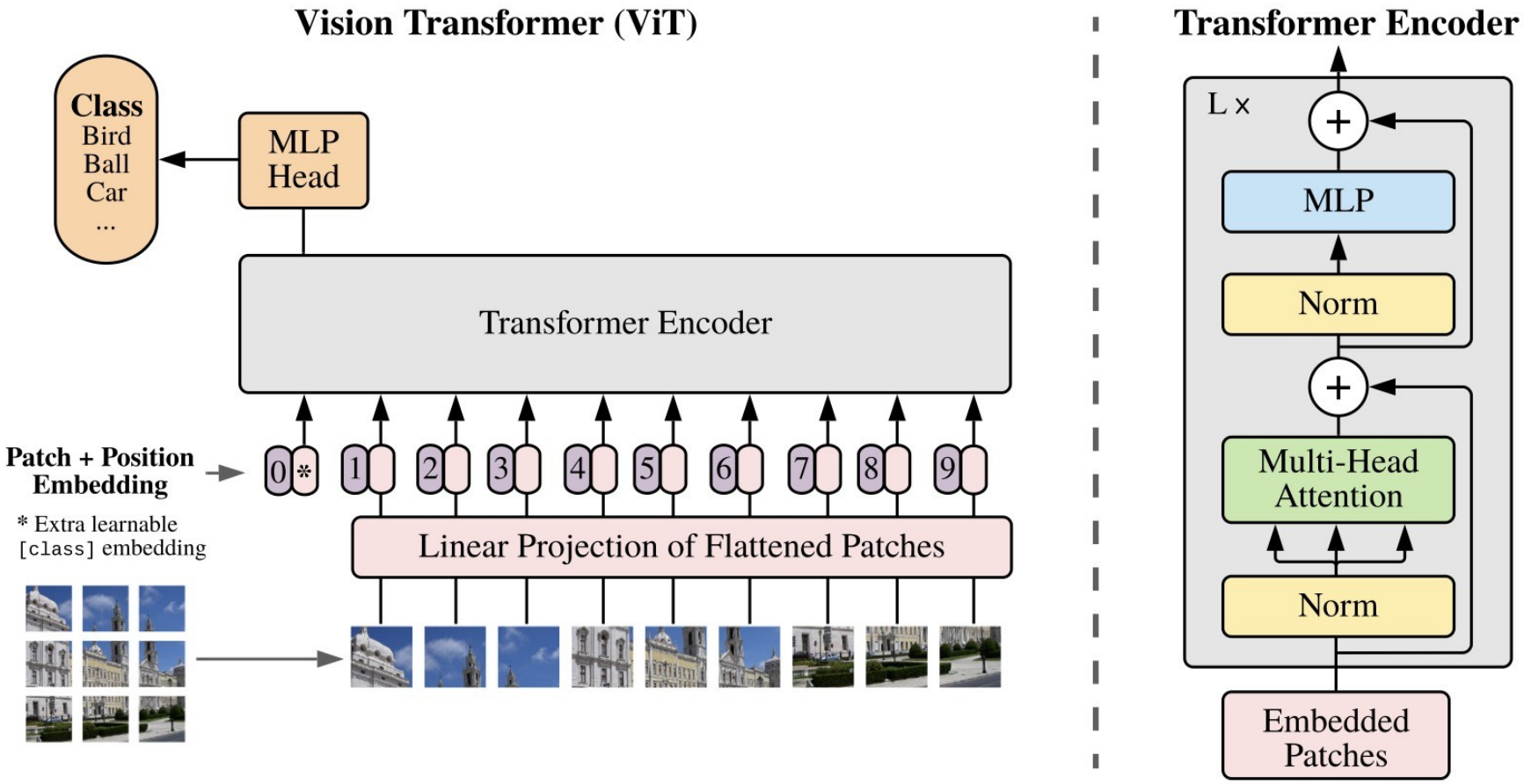

Vision Transformer (ViT)
Class
Bird
Ball
Car
...
MLP Head
Transformer Encoder
Patch + Position Embedding
* Extra learnable [class] embedding
0 * 1 2 3 4 5 6 7 8 9
Linear Projection of Flattened Patches
Transformer Encoder
L x
MLP
Norm
Multi-Head Attention
Norm
Embedded Patches


Linear projection + reshape
Transformer Encoder
Patch + Position Embedding
1
2
3
4
5
6
7
8
9
Linear Projection of Flattened Patches

Original vision transformer

Expensive compute and requires lots of training data

Swin transformer was hierarchical / more efficient for large images



Back to convolution aga

A ConvNet for the 2020s

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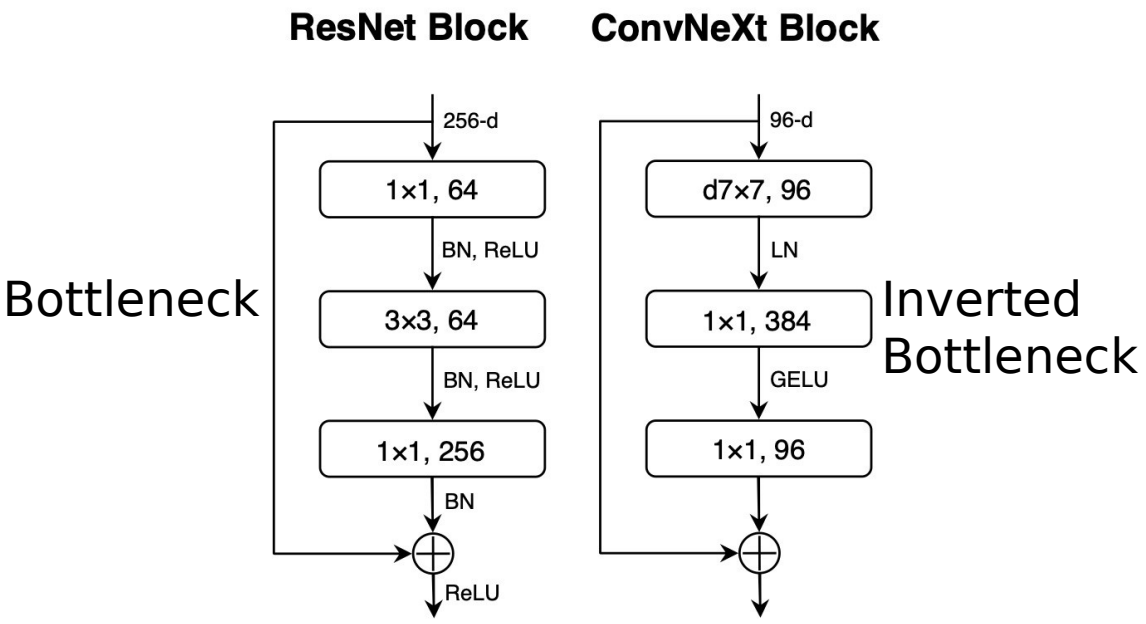
Code: https://github.com/facebookresearch/ConvNeXt

Pure architecture exploration paper, to find a convnet that competes with attention, using attention-inspired modifications:
- larger kernels, (transformers work on large patches)
- layer norm (like transformers),
- Inverted bottleneck layers,
...



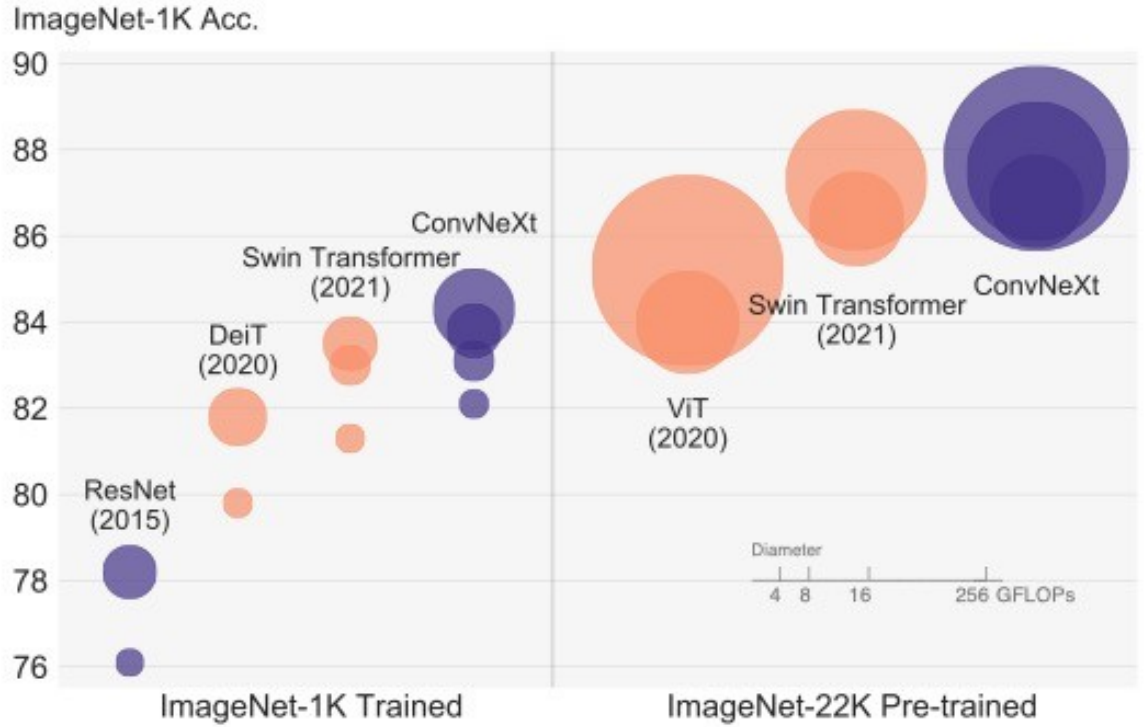
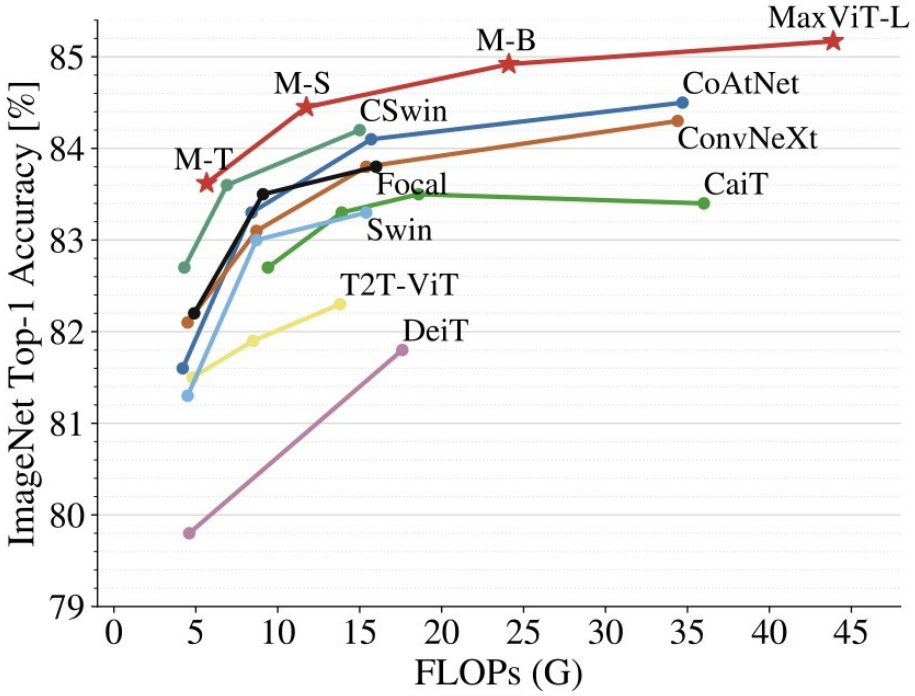
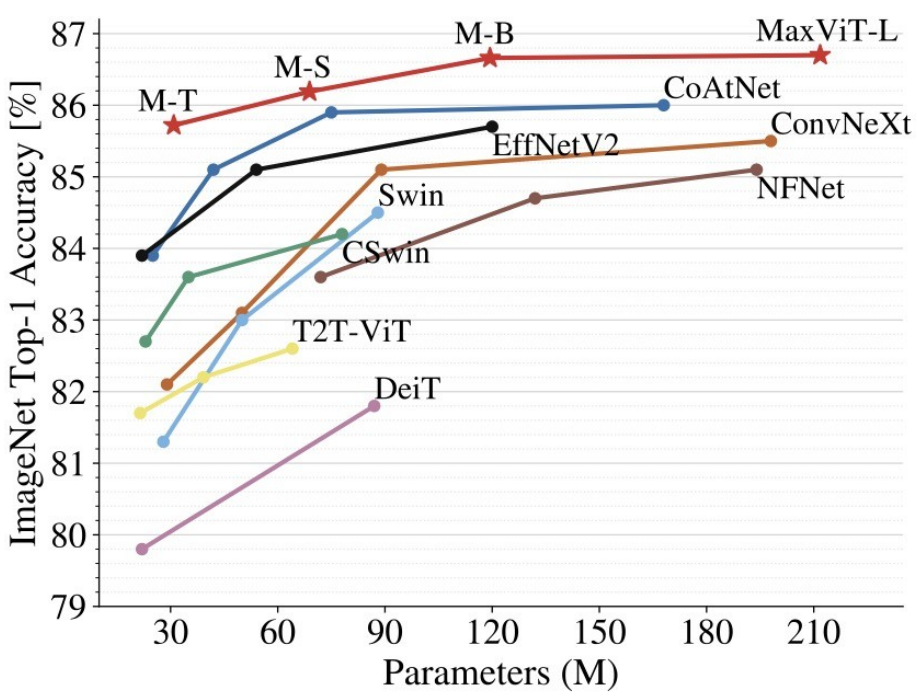


Figure 1. **ImageNet-1K classification** results for • ConvNets and ○ vision Transformers. Each bubble's area is proportional to FLOPs of a variant in a model family. ImageNet-1K/22K models here take 224^2/384^2 images respectively. We demonstrate that a standard ConvNet model can achieve the same level of scalability as hierarchical vision Transformers while being much simpler in design.

Best of both worlds: MaxViT

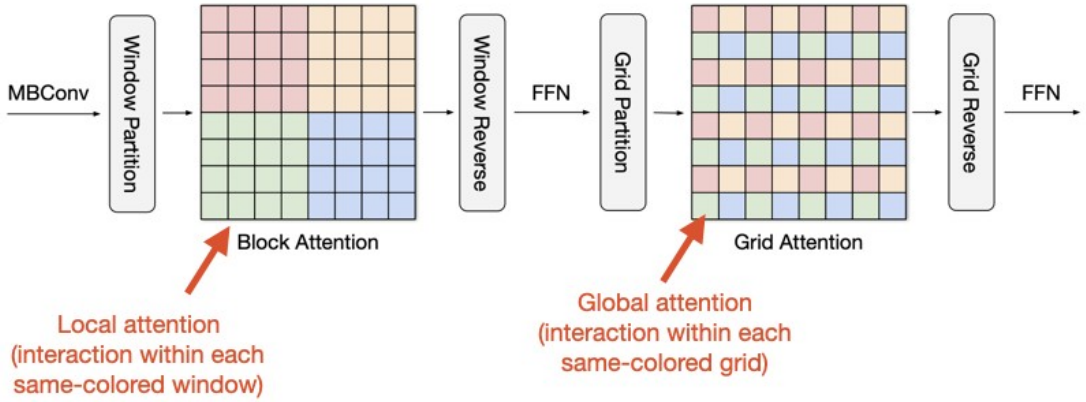


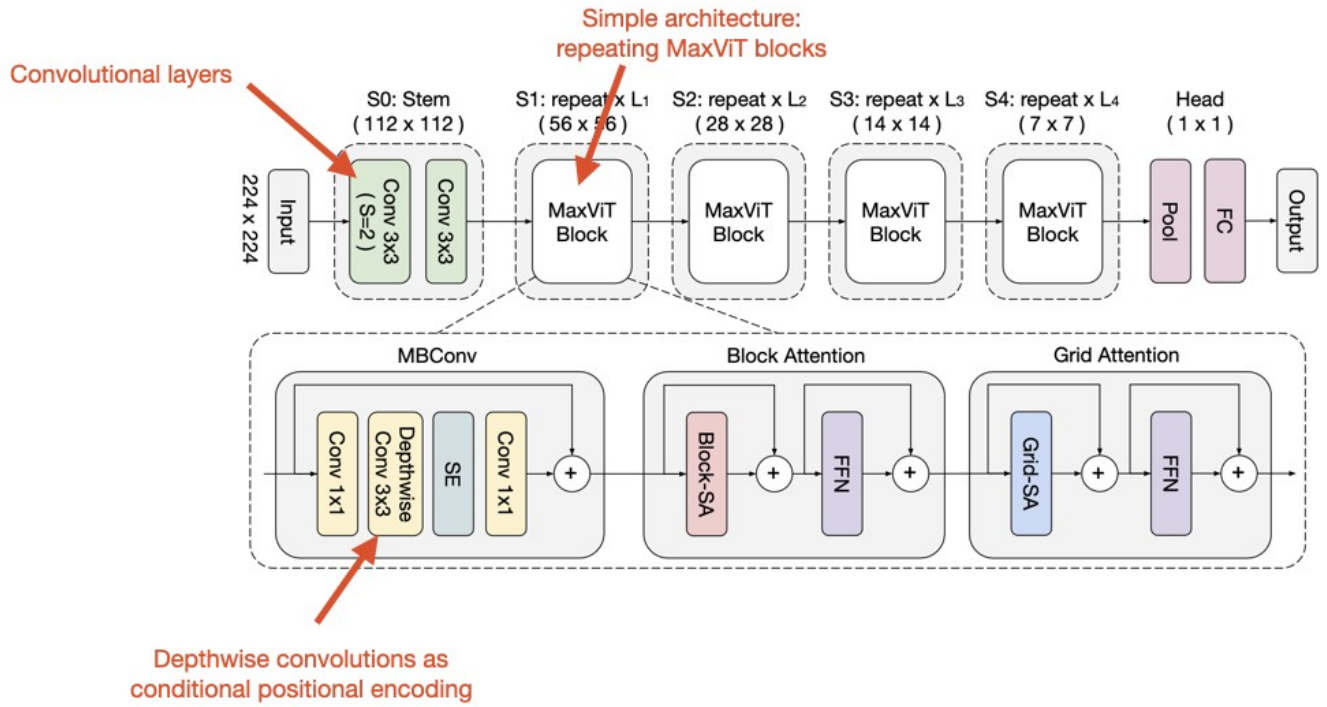
(a) Accuracy *vs.* FLOPs performance scaling curve under ImageNet-1K training setting at input resolution 224×224.



(b) Accuracy *vs.* Parameters scaling curve under ImageNet-1K fine-tuning setting allowing for higher sizes (384/512).

https://arxiv.org/abs/2204.01697 MaxViT





https://arxiv.org/abs/2204.01697 MaxViT

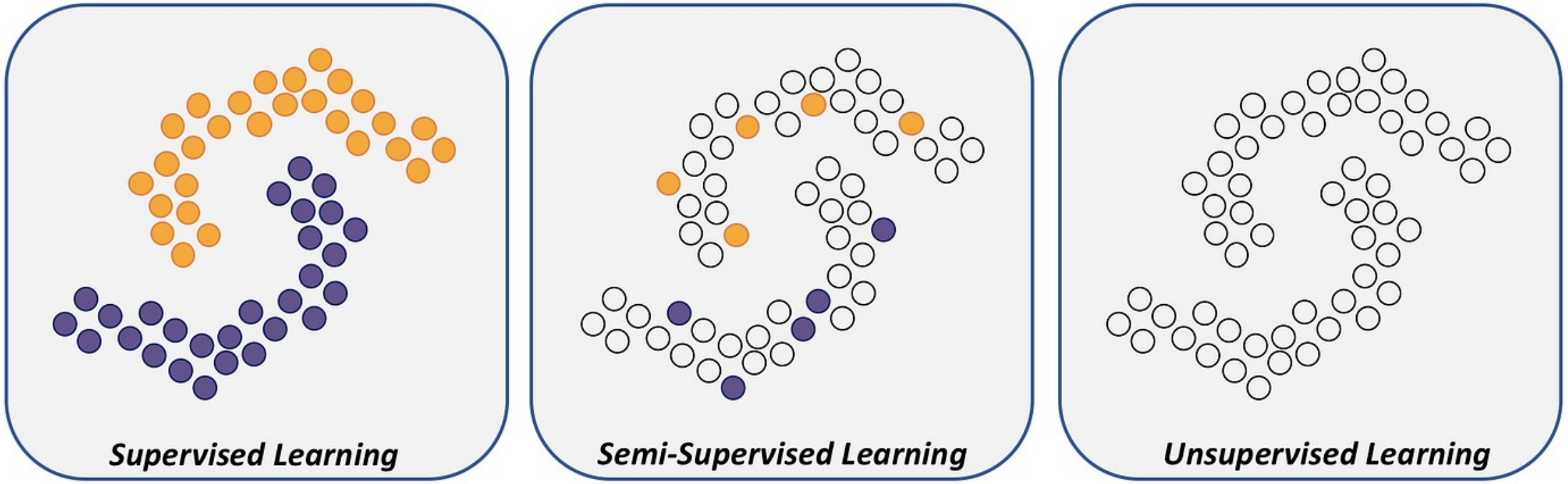
Next time: semi-supervised and self-supervised / contrastive methods in vision

Contrastive style like CLIP/BLIP,

DINO

Multimodal

Semi-supervised Vision



How to exploit the large number of unlabeled examples given in semi-supervised learning?

Transfer learning – Always start here if possible

- Pre-train a good vision backbone on ImageNet

  (Will diffusion models become the ”foundation” for all vision tasks? I think so – seeing evidence in the literature. Arguably CLIP models.)

- Adapt to your new task

Frustratingly Simple Few-Shot Object Detection

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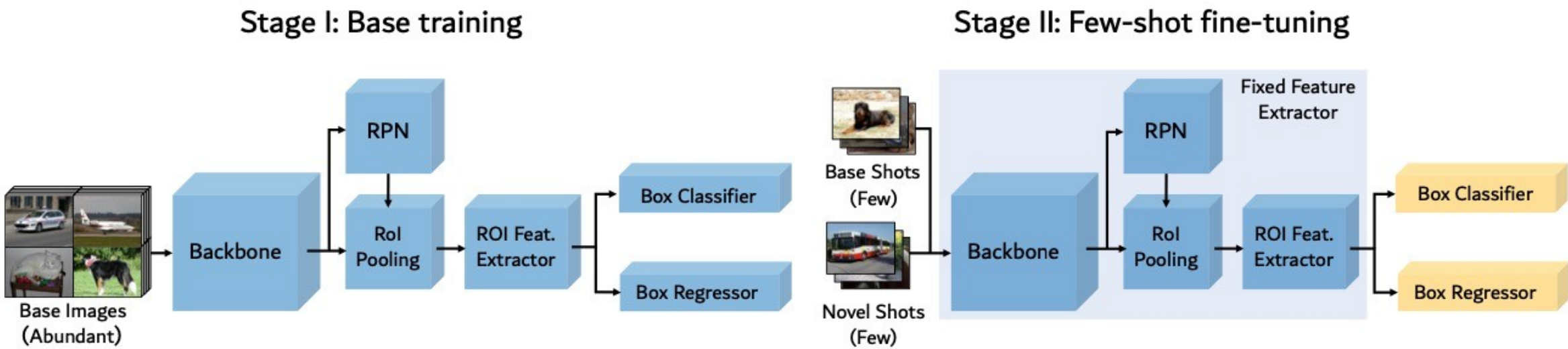

Figure 1. Illustration of our two-stage fine-tuning approach (TFA). In the base training stage, the entire object detector, including both the feature extractor $\mathcal{F}$ and the box predictor, are jointly trained on the base classes. In the few-shot fine-tuning stage, the feature extractor components are fixed and only the box predictor is fine-tuned on a balanced subset consisting of both the base and novel classes.

A BASELINE FOR FEW-SHOT IMAGE CLASSIFICATION

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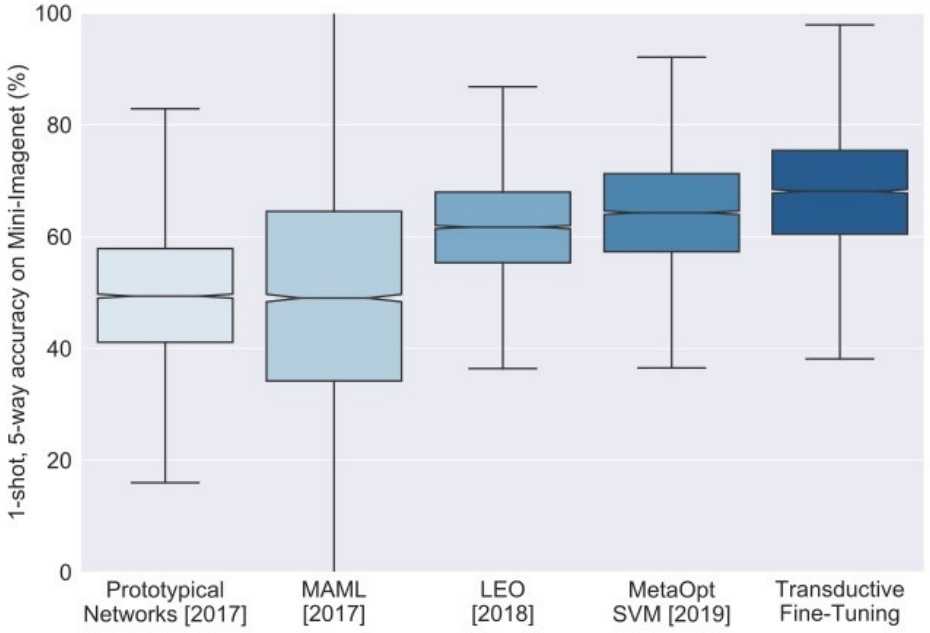

Figure 1: **Are we making progress?** The box-plot illustrates the performance of state-of-the-art few-shot algorithms on the Mini-ImageNet (Vinyals et al., 2016) dataset for the 1-shot 5-way protocol. The boxes show the $\pm$ 25% quantiles of the accuracy while the notches indicate the median and its 95% confidence interval. Whiskers denote the 1.5$\times$ interquartile range which captures 99.3% of the probability mass for a normal distribution. The spread of the box-plots are large, indicating that the standard deviations of the few-shot accuracies is large too. This suggests that progress may be illusory, especially considering that none outperform the simple transductive fine-tuning baseline discussed in this paper (rightmost).

Pre-train, then fine-tune on target

3.2 TRANSDUCTIVE FINE-TUNING

In (4), we assumed that there is a single query sample. However, we can also process multiple query samples together, and perform the minimization over all unknown query labels. We introduce a regularizer, similar to Grandvalet & Bengio (2005), as we seek outputs with a peaked posterior, or low Shannon Entropy $\mathbb{H}$. So the transductive fine-tuning phase solves for

$$\Theta^* = \arg\min_{\Theta} \frac{1}{N_s} \sum_{(x,y)\in\mathcal{D}_s} -\log p_\Theta\left(y \mid x\right) + \frac{1}{N_q} \sum_{(x,y)\in\mathcal{D}_q} \mathbb{H}(p_\Theta(\cdot \mid x)). \tag{8}$$

$$\frac{1}{N_q} \sum_{(x,y)\in\mathcal{D}_q} \mathbb{H}(p_\Theta(\cdot \mid x))$$

Transductive fine-tuning also tries to give low entropy to unlabeled examples